

Lecture 4:

Ablation

Firn densification

Calving

Sub-shelf melting

Surface energy balance

- components

- controls

- how it relates to SMB

Discretization and regional climate modelling (prep. for the practical on Tuesday).

Firn densification processes

Its complicated!

Settling

movement of grains relative to each other
increases density up to $\sim 550 \text{ kg m}^{-3}$

Sublimation

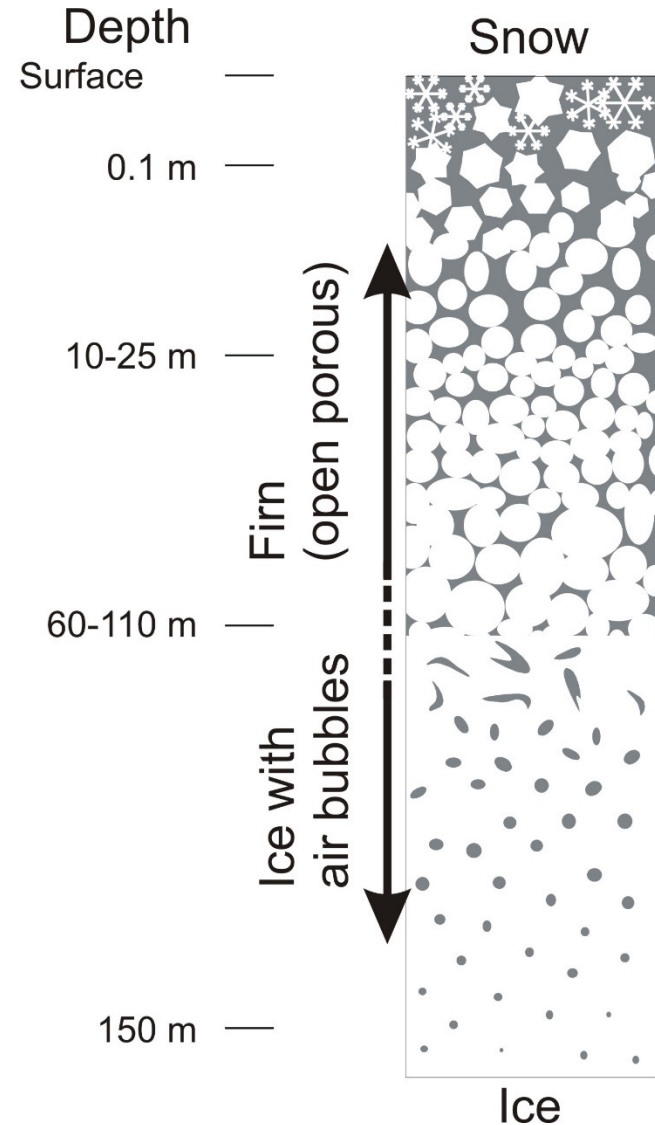
$> 550 \text{ kg m}^{-3}$,
movement of molecules to 'necks' of grains due
to vapour-pressure gradients in pores.

Deformation of grains

Through diffusion creep
(more details next week)

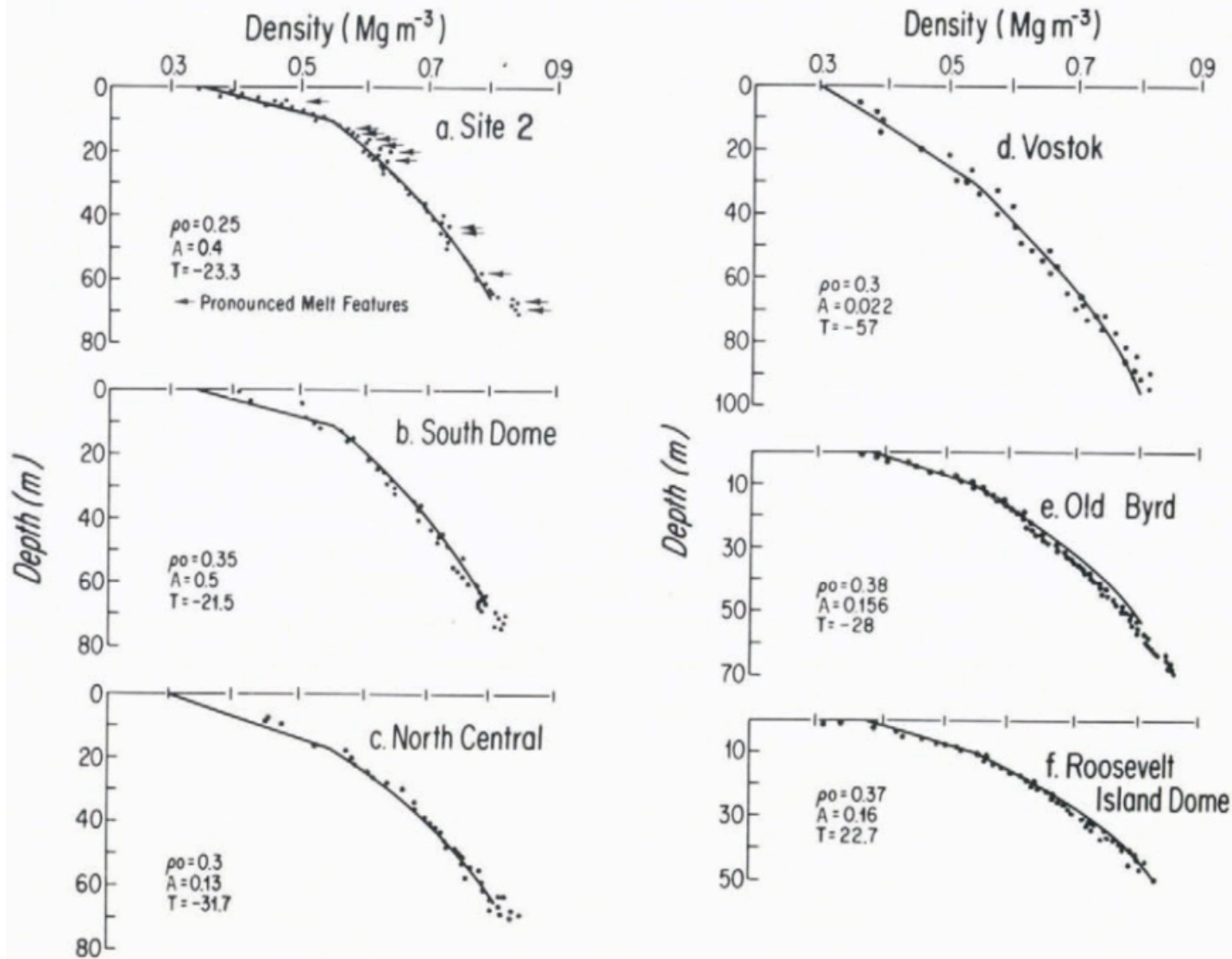
At $\sim 830 \text{ kg m}^{-3}$ pores "close off"
and firn becomes ice.

Compression of air in bubbles allows ice density to
reach $\sim 917 \text{ kg m}^{-3}$.



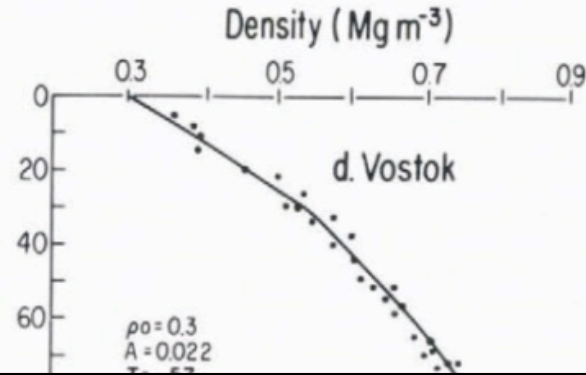
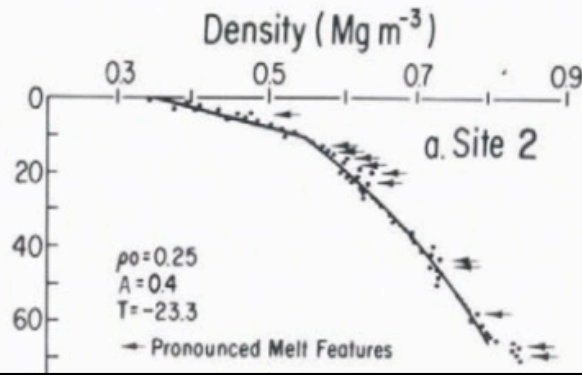
University of Copenhagen, Center for Ice and Climate

An empirical model



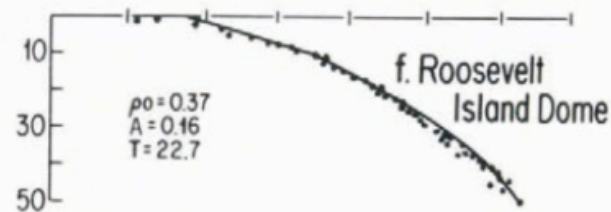
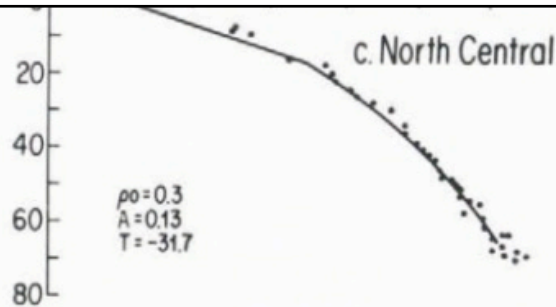
Herron and Langway (1980)

An empirical model



Seems to work, but very little physics in there and it needs tuning each time!

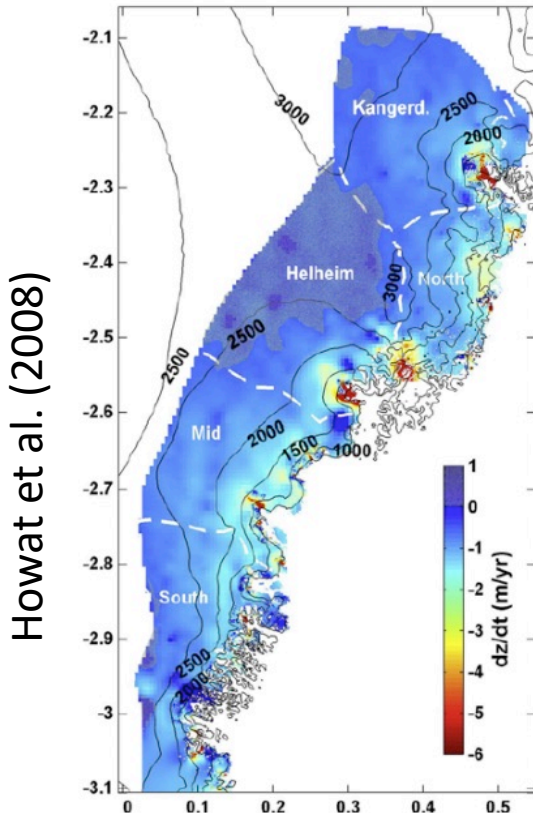
Not very useful for prediction.



Herron and Langway (1980)

Why is getting this right important?

1. Geodetic mass balance



2. Ice core paleo-climate records

Ice cores preserve old ice and old air.

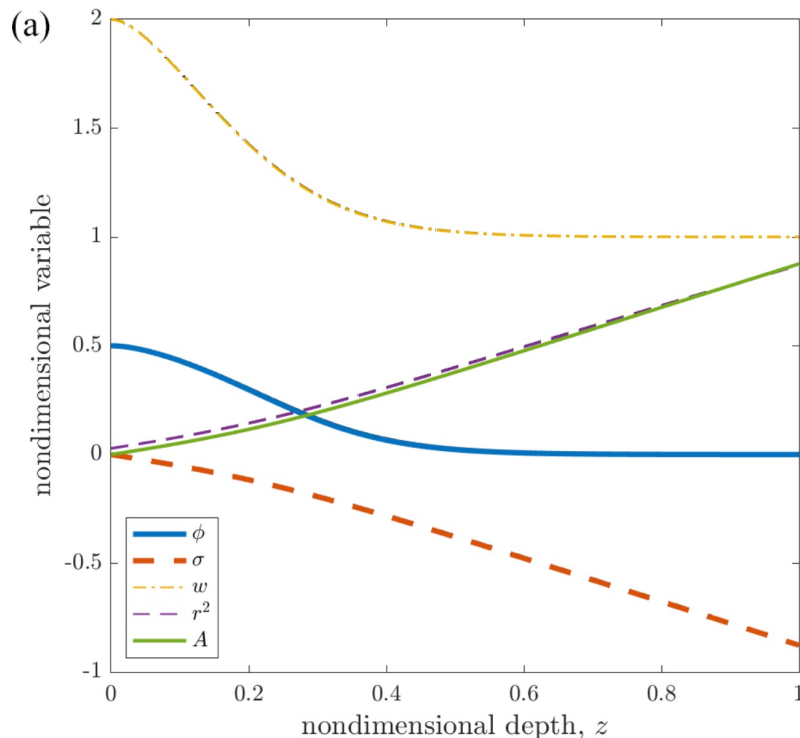
The ice is older than the air by an amount that depends on firn densification rates.



Grain-size evolution controls the accumulation dependence of modelled firn thickness

Jonathan Kingslake, Robert Skarbek, Elizabeth Case, and Christine McCarthy

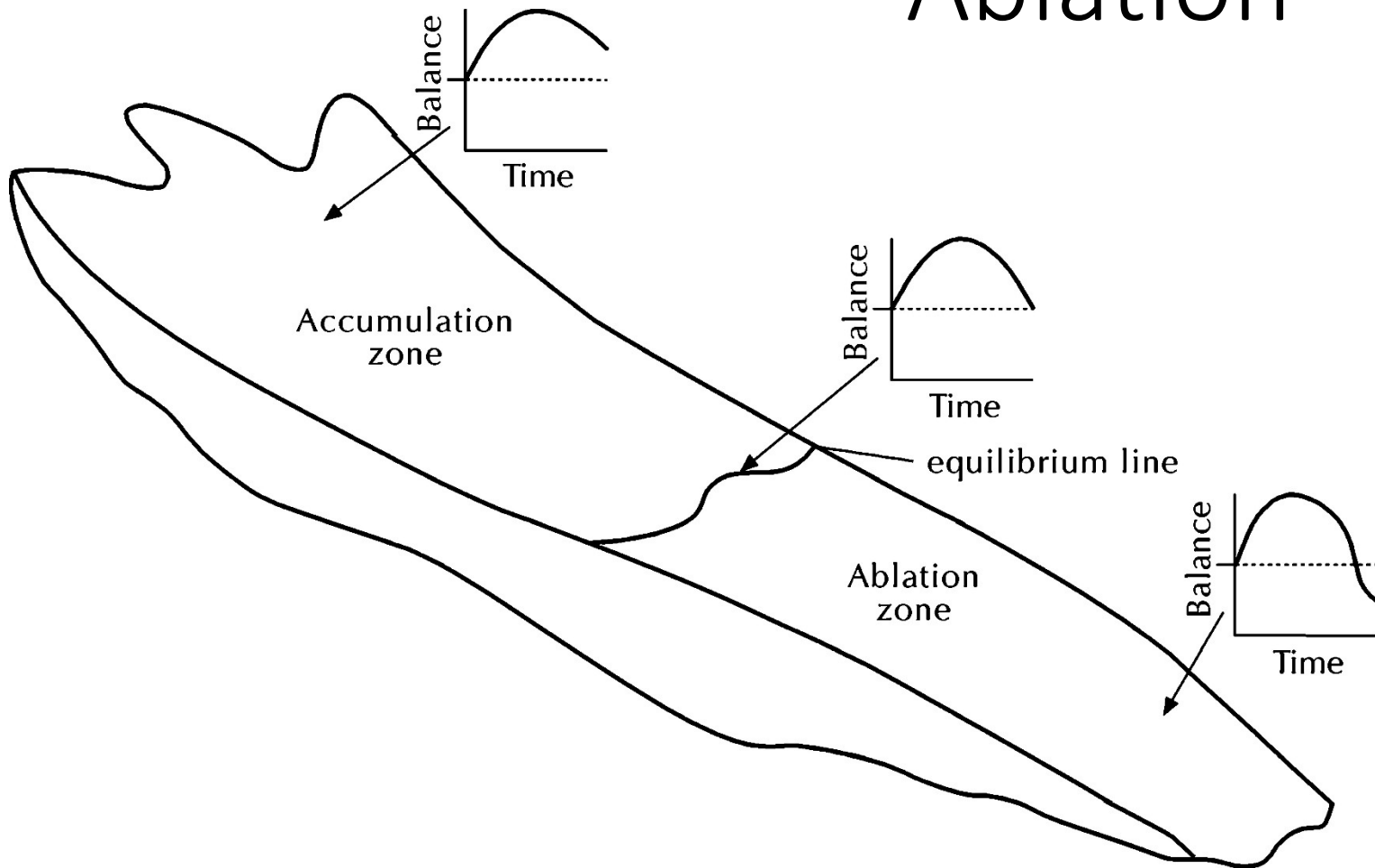
Lamont-Doherty Earth Observatory, Department of Earth and Environmental Science,
Columbia University, New York, NY 10027, USA



$$\frac{\partial \sigma}{\partial z} = -\rho_i g (1 - \phi),$$
$$\sigma(0) = 0,$$

The Cryosphere, 16, 3413–3430, 2022
<https://doi.org/10.5194/tc-16-3413-2022>

Ablation



Ablating surfaces: 1. Calving front



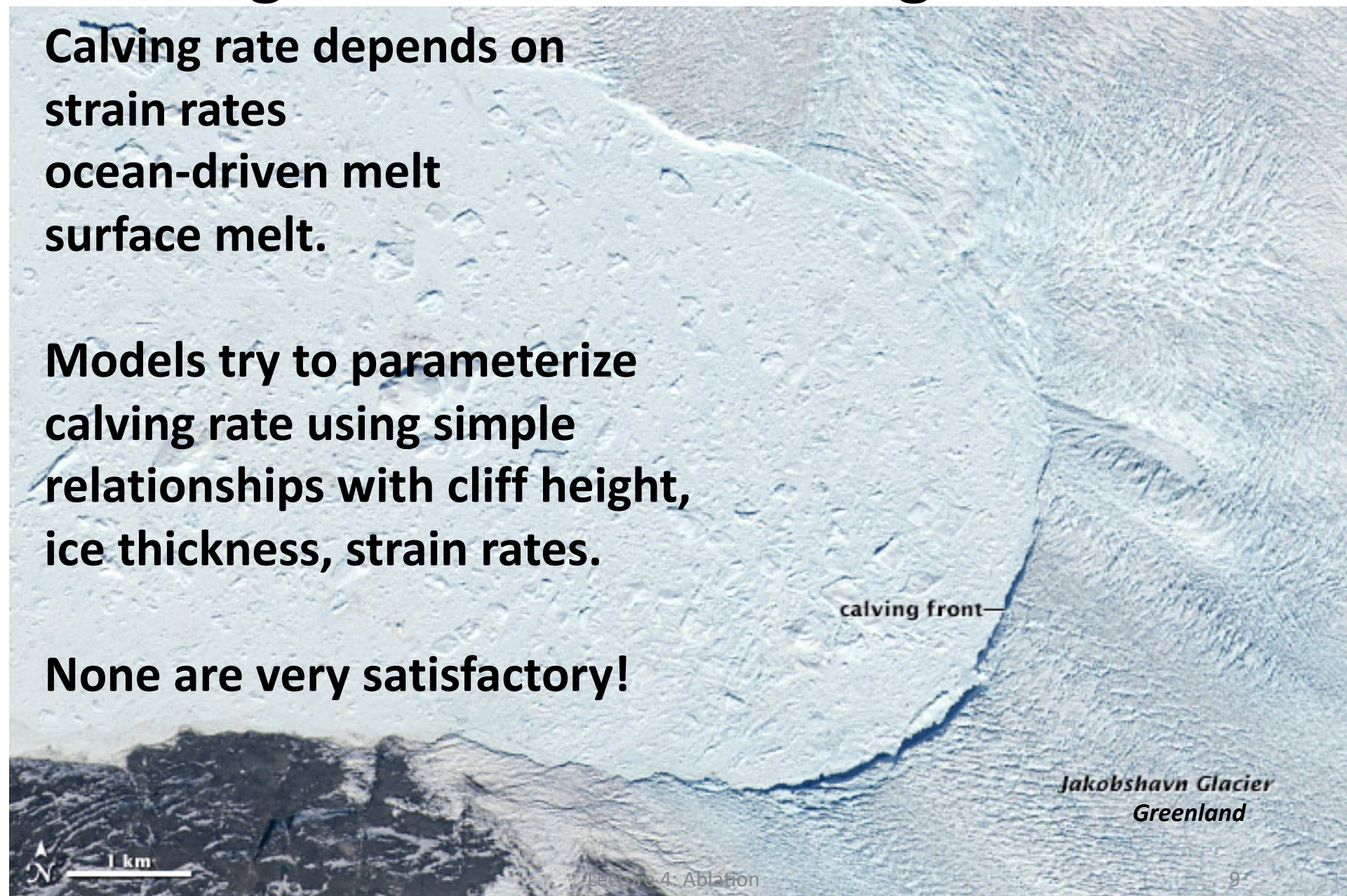
Columbia Glacier, Alaska

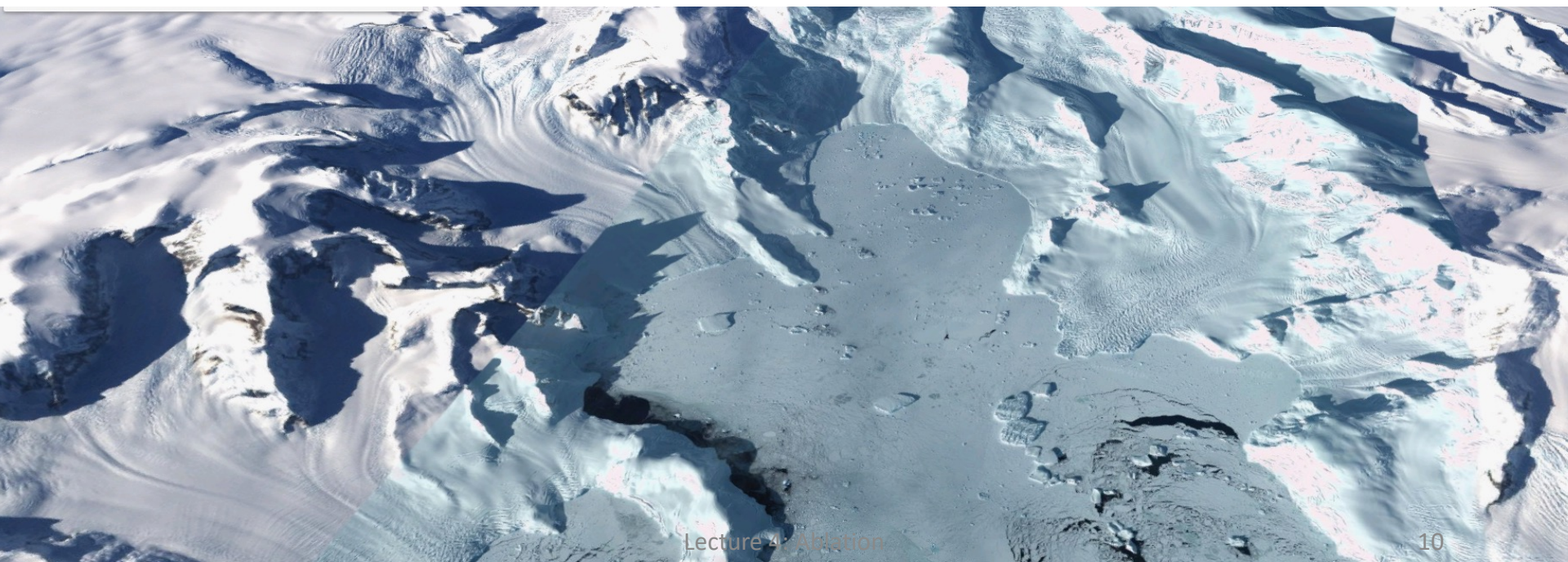
Ablating surfaces: 1. Calving front

**Calving rate depends on
strain rates
ocean-driven melt
surface melt.**

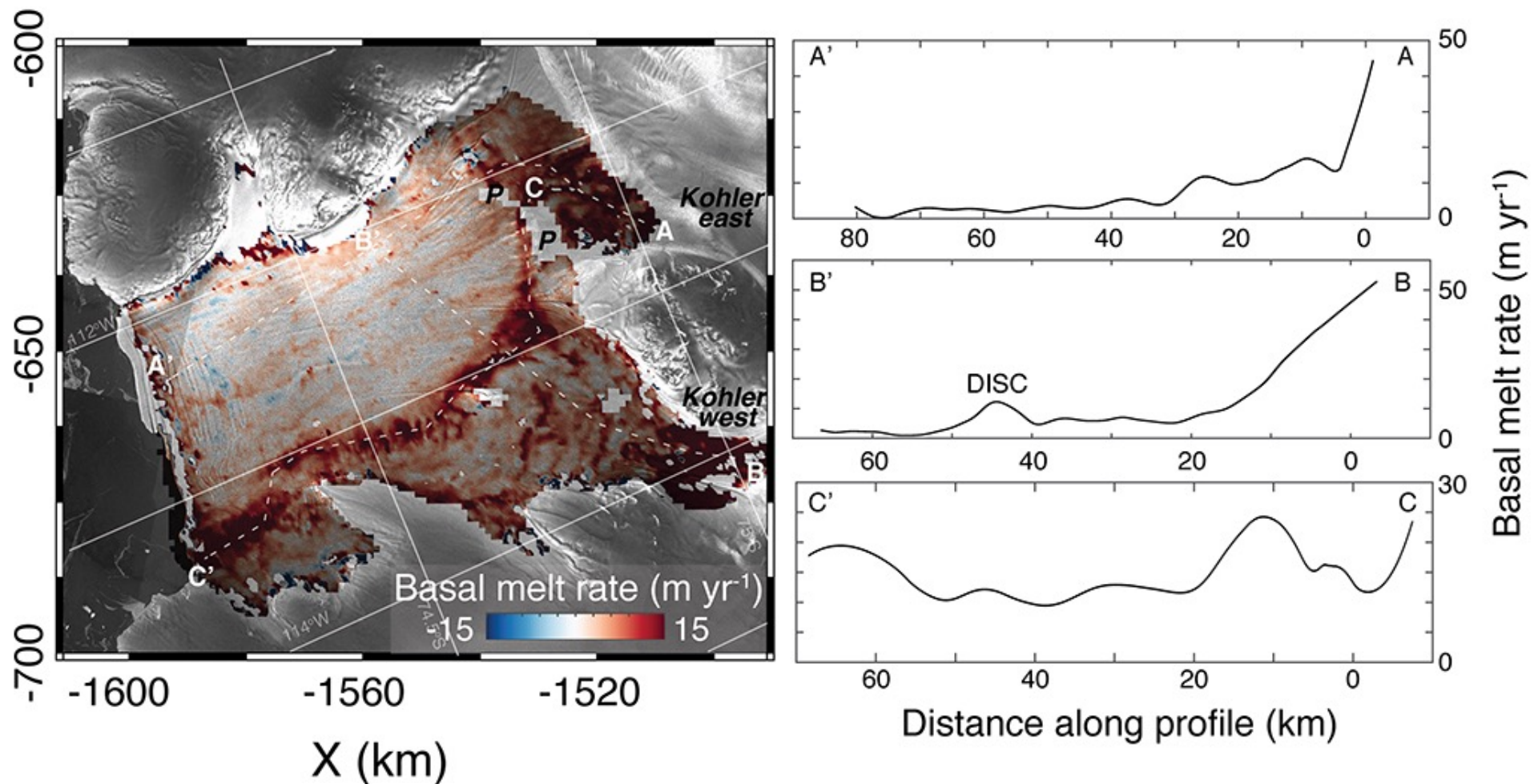
**Models try to parameterize
calving rate using simple
relationships with cliff height,
ice thickness, strain rates.**

None are very satisfactory!





Ablating surfaces: 2. Base



Gourmelen et al. (2017)

Ablating surfaces: 2. Base

Measuring basal melt

Mass conservation leads to:

$$\frac{dH}{dt} = -\frac{dHu}{dx} + \frac{\dot{b}_s}{\rho_s} + \frac{\dot{b}_b}{\rho_i}$$

H is ice thickness,
 u is ice flow speed
 $\dot{b}_s$ is surface mass balance
 $\dot{b}_b$ is basal mass balance

Ablating surfaces: 2. Base

Measuring basal melt

Mass conservation leads to:

We measure these

Measure or model these

$$\frac{dH}{dt} = -\frac{dHu}{dx} + \frac{\dot{b}_s}{\rho_s} + \frac{\dot{b}_b}{\rho_i}$$

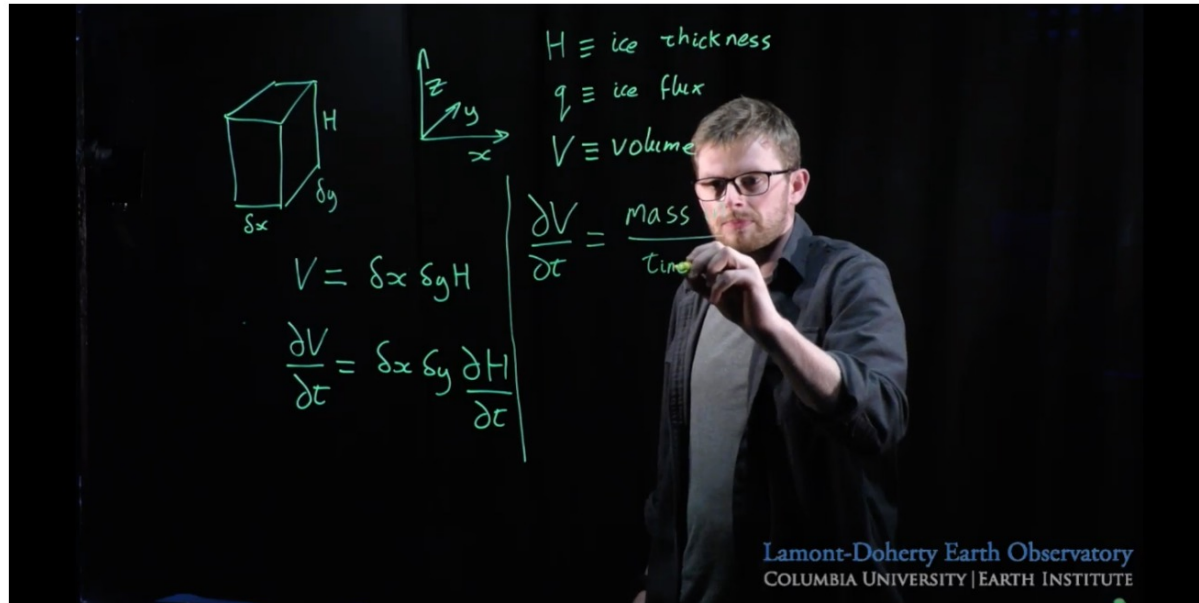
to compute this.

H is ice thickness,
 u is ice flow speed
 $\dot{b}_s$ is surface mass balance
 $\dot{b}_b$ is basal mass balance

Ablating surfaces: 2. Base

Measuring basal melt

Mass cor

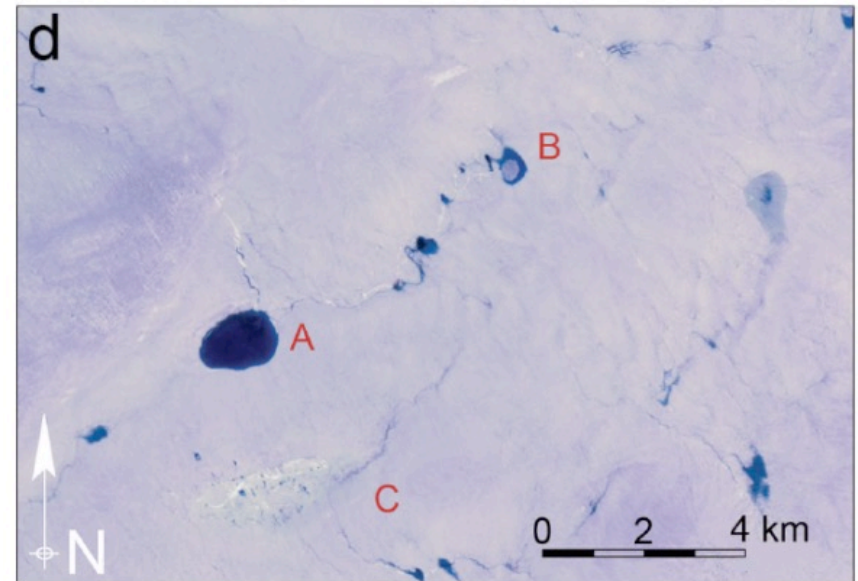
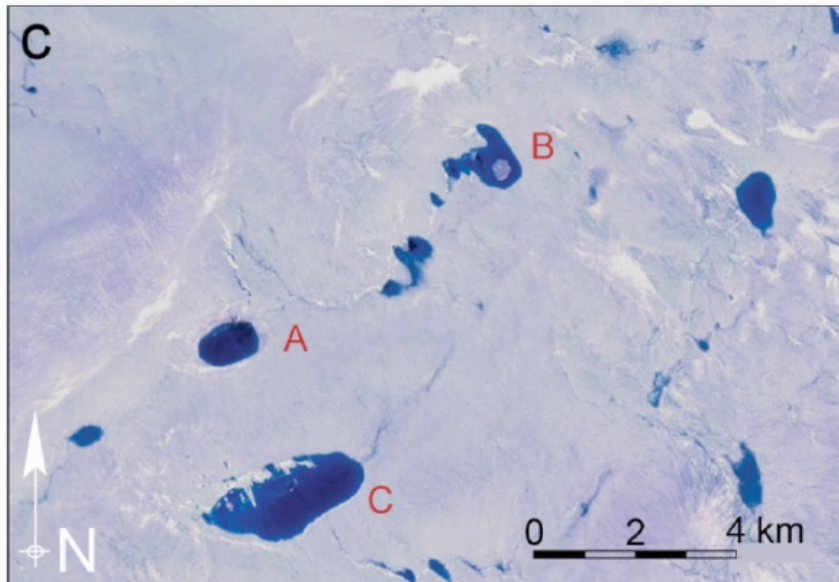
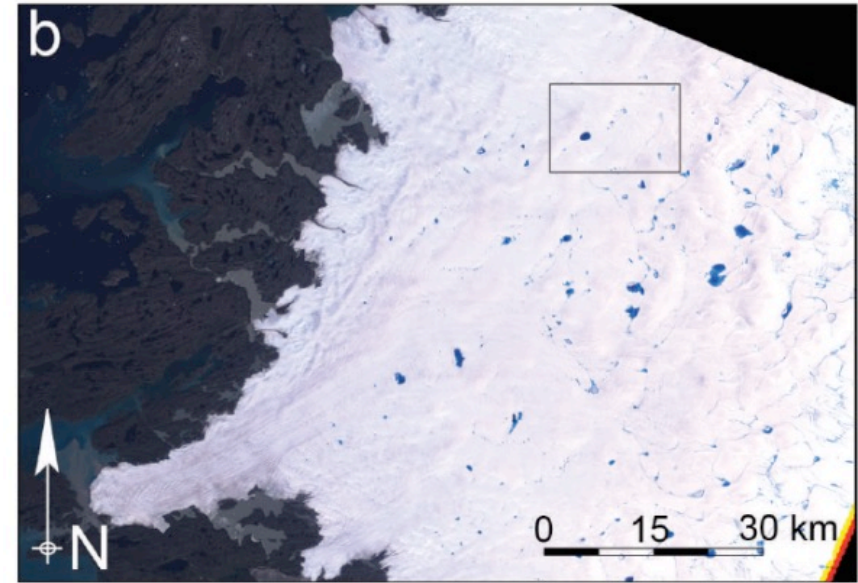
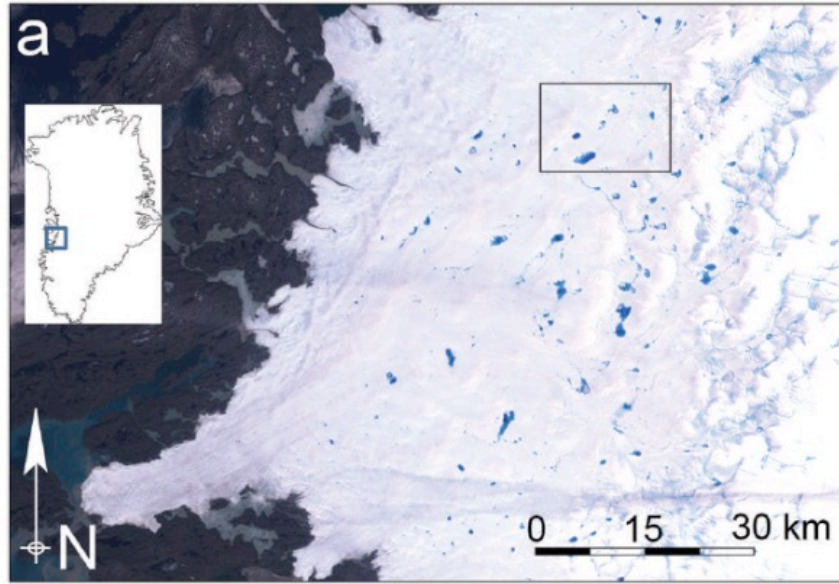


compute

H is ice thickness,
 u is ice flow speed
 $\dot{b}_s$ is surface mass balance
 $\dot{b}_b$ is basal mass balance

<https://Ideo-it.Ideo.columbia.edu/content/lightboard-studio-lamont>

Ablating surfaces: 3. Surface

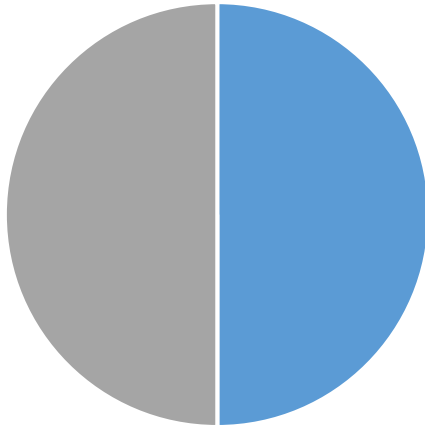


Ablating surfaces: 3. Surface



Very approximately:

Greenland



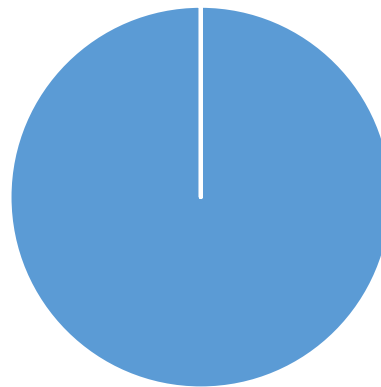
■ Surface ■ Base ■ Calving

Antarctica



■ Surface ■ Base ■ Calving

Land-terminating mountain glaciers



■ Surface ■ Base ■ Calving

Surface energy balance

$$E_N = E_s^{\downarrow} + E_s^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_G + E_H + E_E + E_P$$

Surface energy balance

$$E_N = E_s^{\downarrow} + E_s^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + \cancel{E_G} + E_H + E_E + \cancel{E_P}$$

Surface energy balance

$$E_N = E_S^{\downarrow} + E_S^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_H + E_E$$

Surface energy balance

$$E_N = E_S^{\downarrow} + E_S^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_H + E_E$$

Radiative fluxes

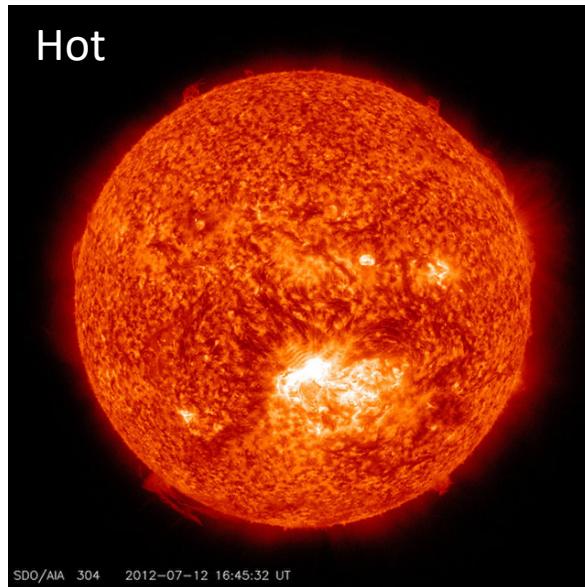
Black-body radiation

$$E = \sigma T^4$$

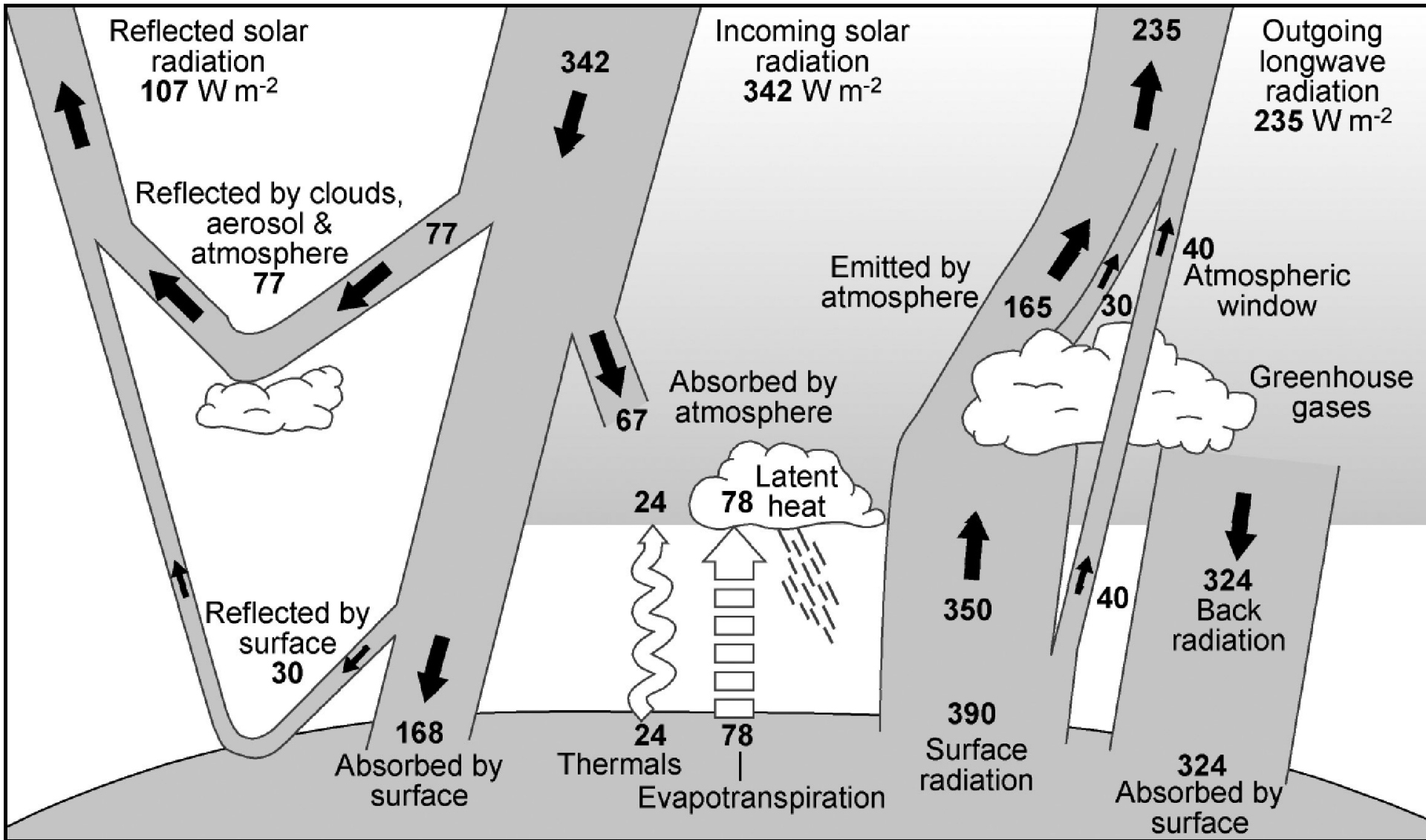
Stefan-Boltzmann
equation

$$\lambda_{max} = \frac{b}{T}$$

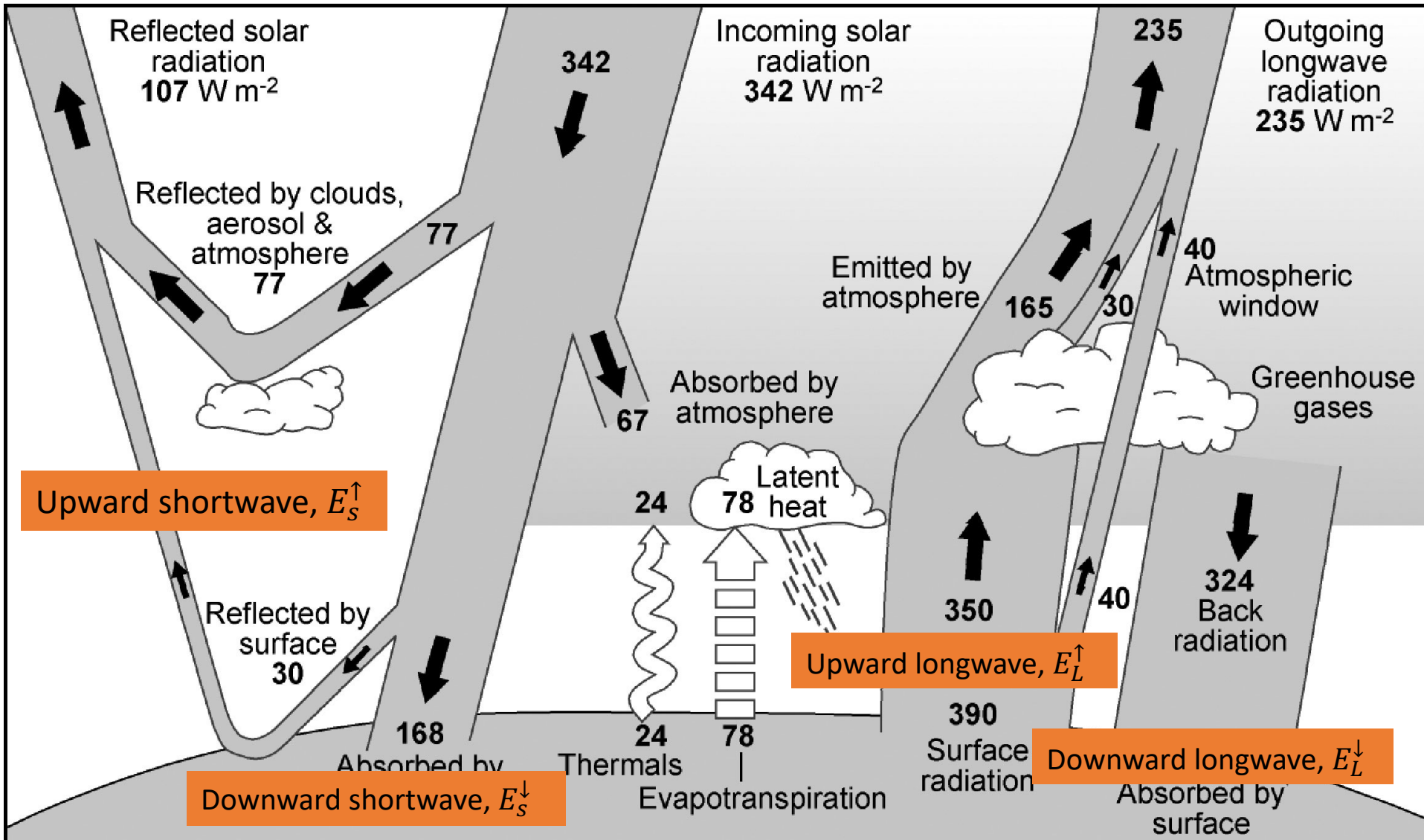
Wein's
displacement law

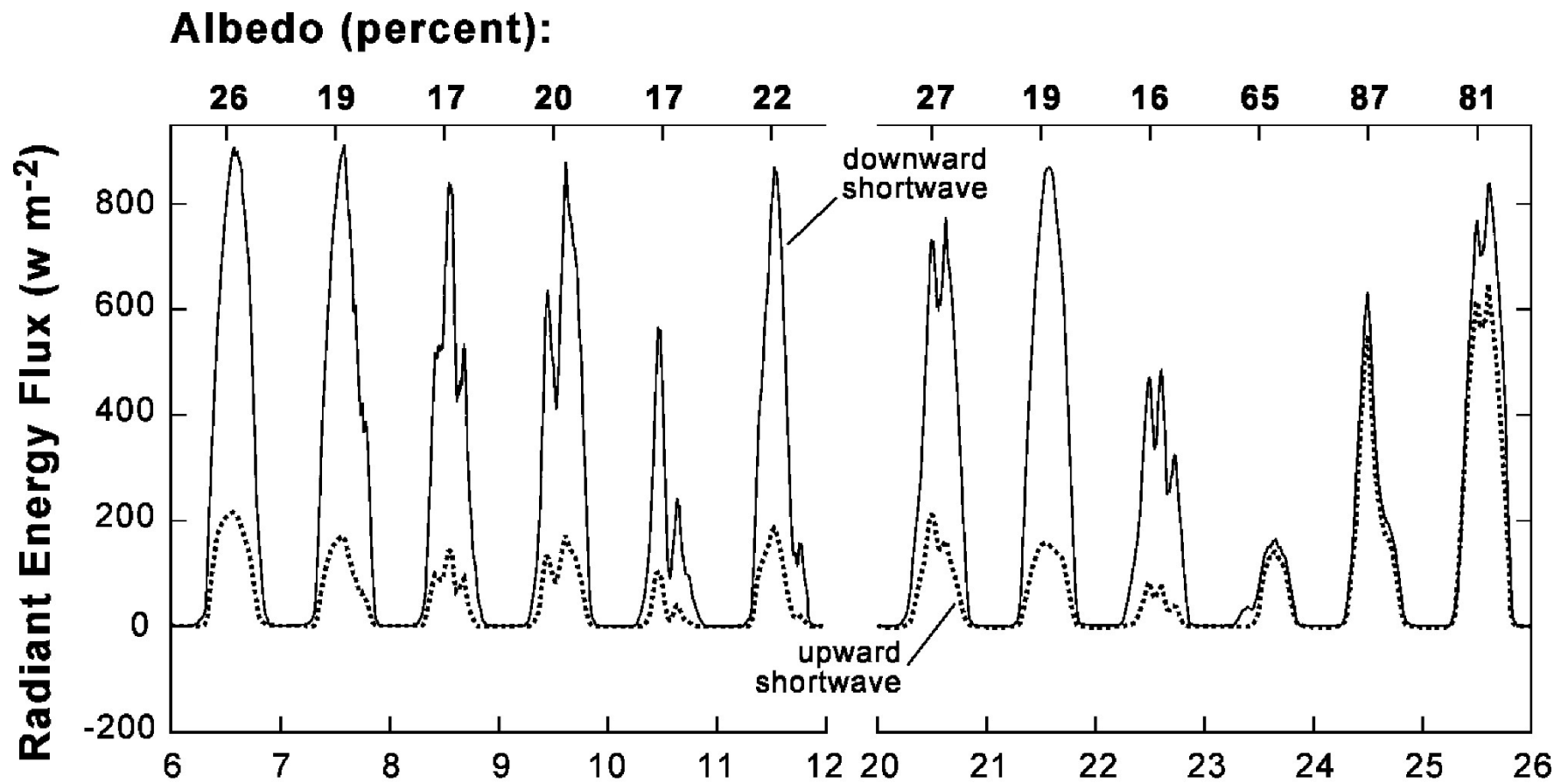


Radiative fluxes in the atmosphere

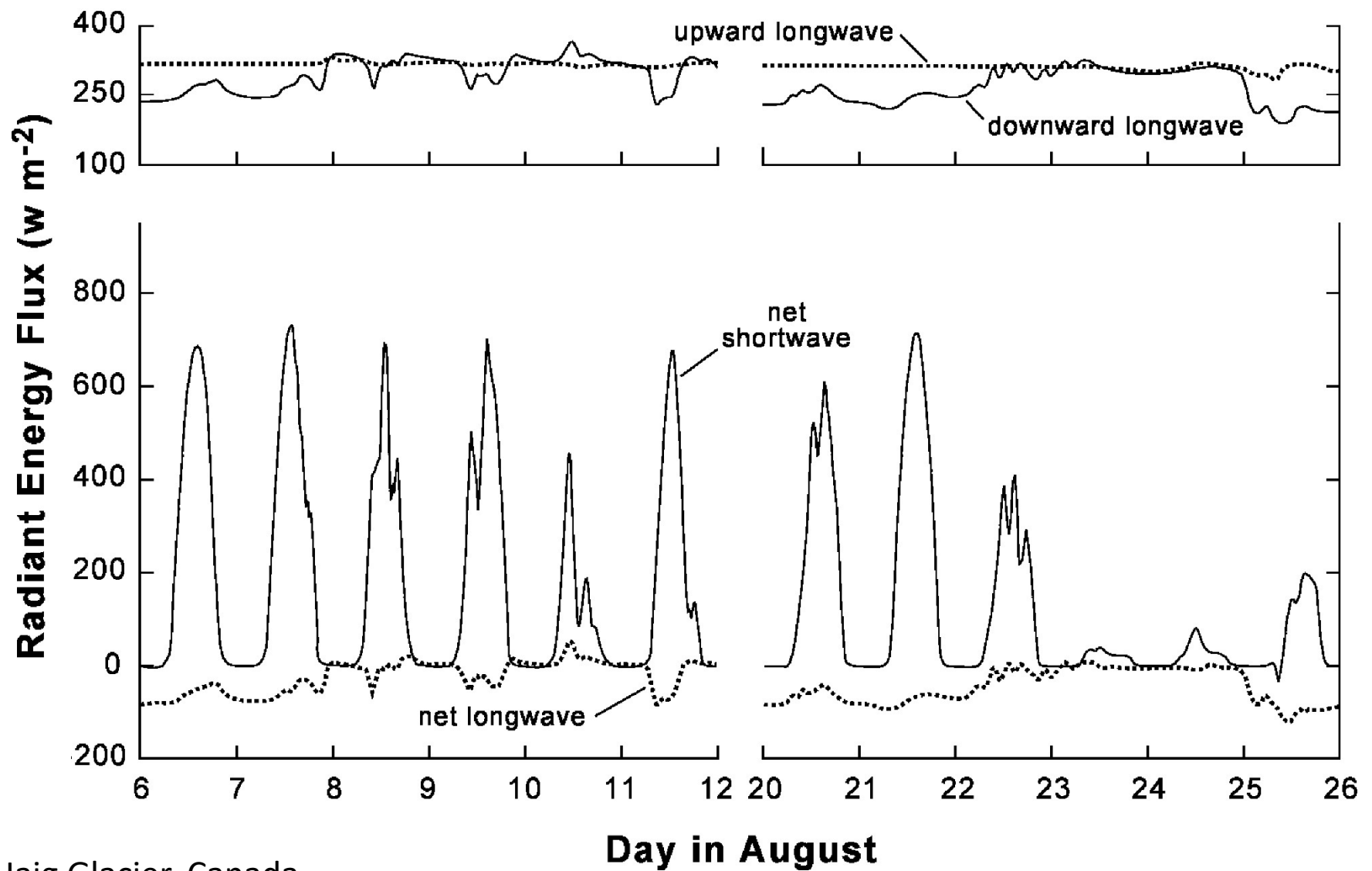


Radiative fluxes in the atmosphere



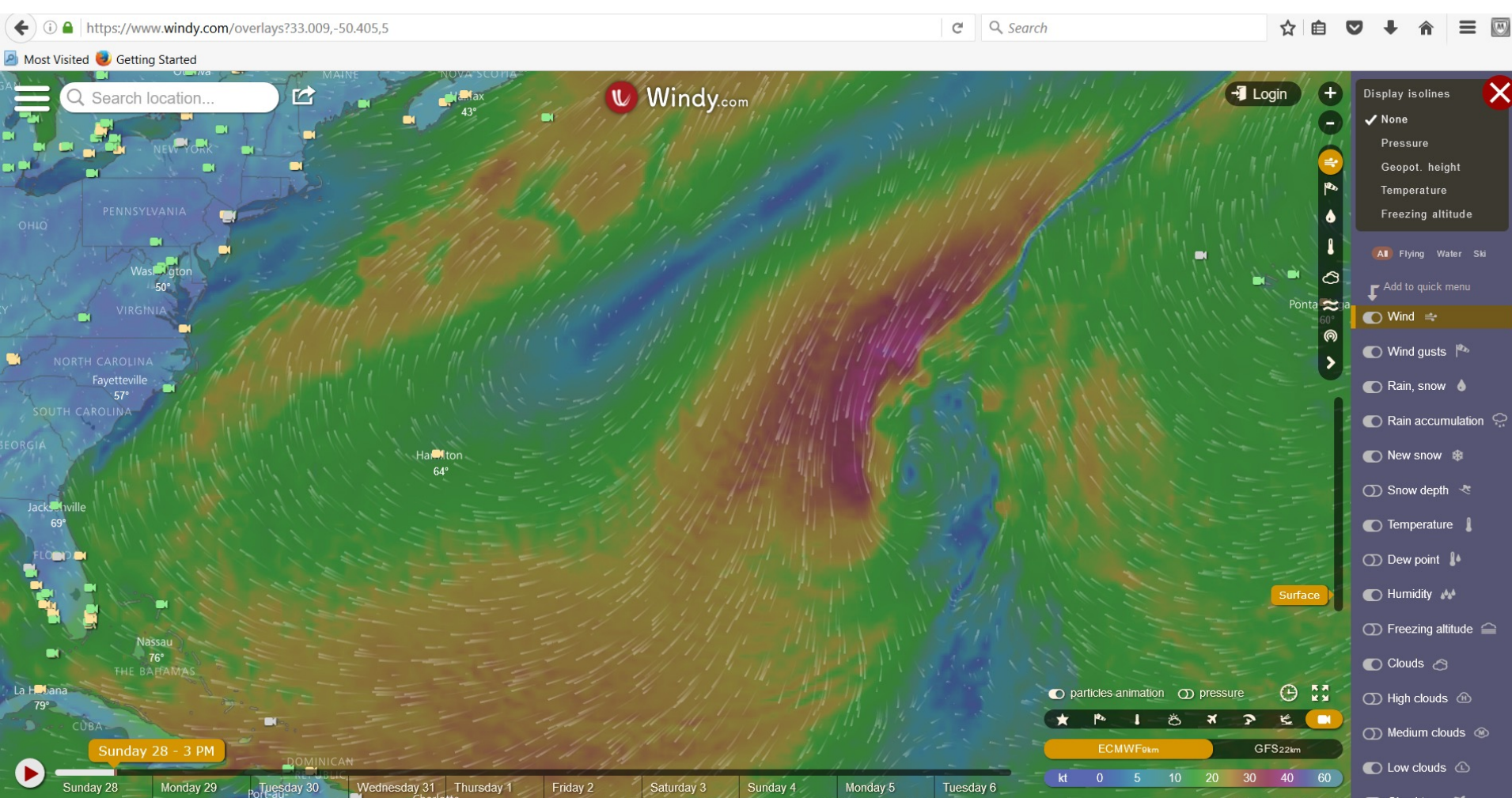


Haig Glacier, Canada,
Cuffey and Paterson (2010)



Haig Glacier, Canada,
Cuffey and Paterson (2010)

Turbulent fluxes



<https://www.windy.com/overlays?32.990,-50.405,5>

Surface energy balance

$$E_N = E_S^{\downarrow} + E_S^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_H + E_E$$

Turbulent fluxes

Depend sensitively on weather conditions throughout the atmosphere,

Surface energy balance

$$E_N = E_S^{\downarrow} + E_S^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_H + E_E$$

Turbulent fluxes

Sensible heat flux

$$E_H \propto u(T_a - T_s)$$

Latent heat flux

$$E_E \propto u(q_a - q_s)$$

Depend sensitively on weather conditions throughout the atmosphere, but these eqns use conditions just above the surface.

where,
 u is wind speed,
 T_a is air temperature
 T_s is surface temperature
 q_a is humidity of the air
 q_s is humidity at the surface

Surface energy balance

$$E_N = E_S^\downarrow + E_S^\uparrow + E_L^\downarrow + E_L^\uparrow + E_H + E_E$$

Turbulent fluxes

Sensible heat flux

$$E_H \propto u(T_a - T_s)$$

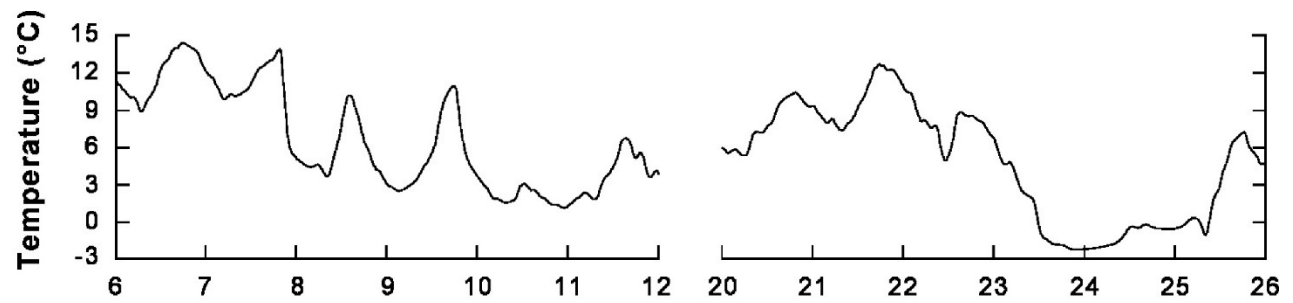
Latent heat flux

$$E_E \propto u(q_a - q_s)$$

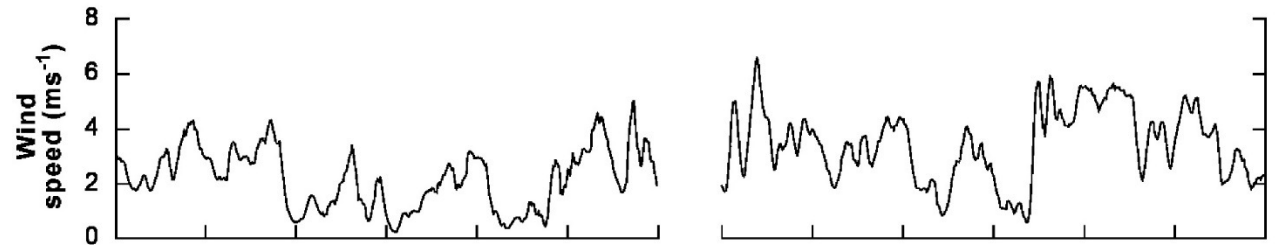
Corresponds to a mass exchange that can be positive (deposition) or negative (sublimation/evaporation).

U is wind speed,
 T_a is air temperature
 T_s is surface temperature
 q_a is humidity of the air
 q_s is humidity at the surface

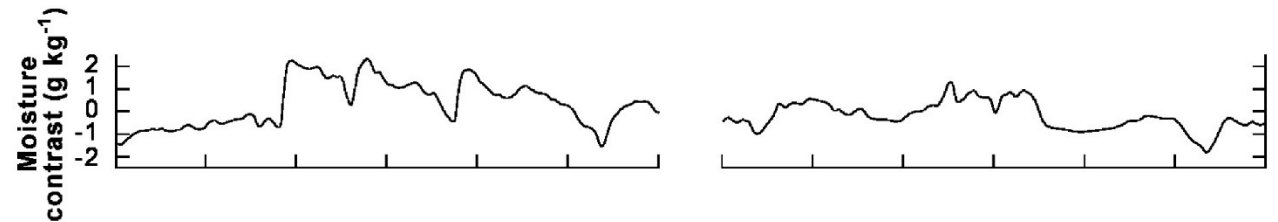
T_a



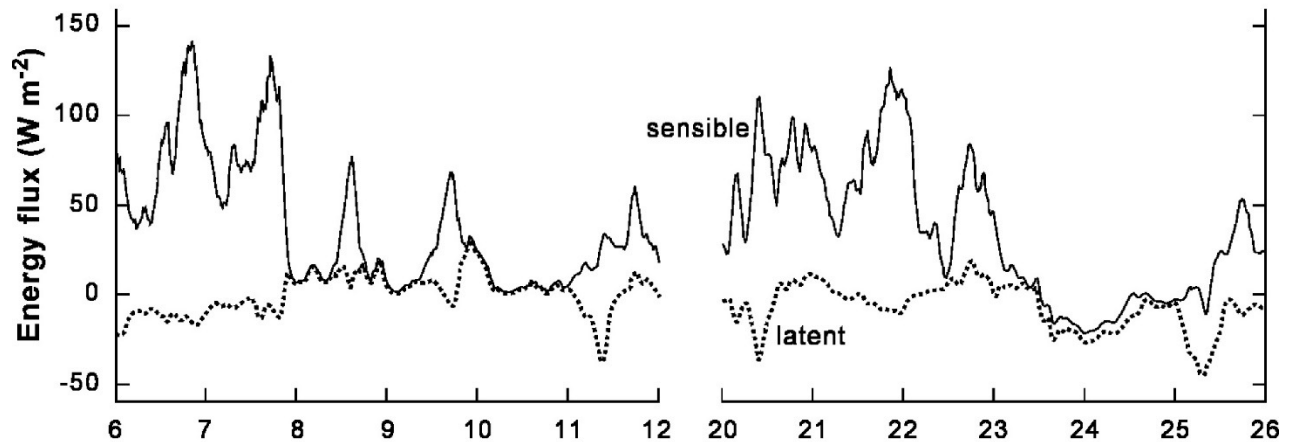
U



$q_a - q_s$



E_H E_E



Haig Glacier, Canada,
 Cuffey and Paterson (2010)

Day in August

$$E_N = E_S^{\downarrow} + E_S^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_H + E_E$$

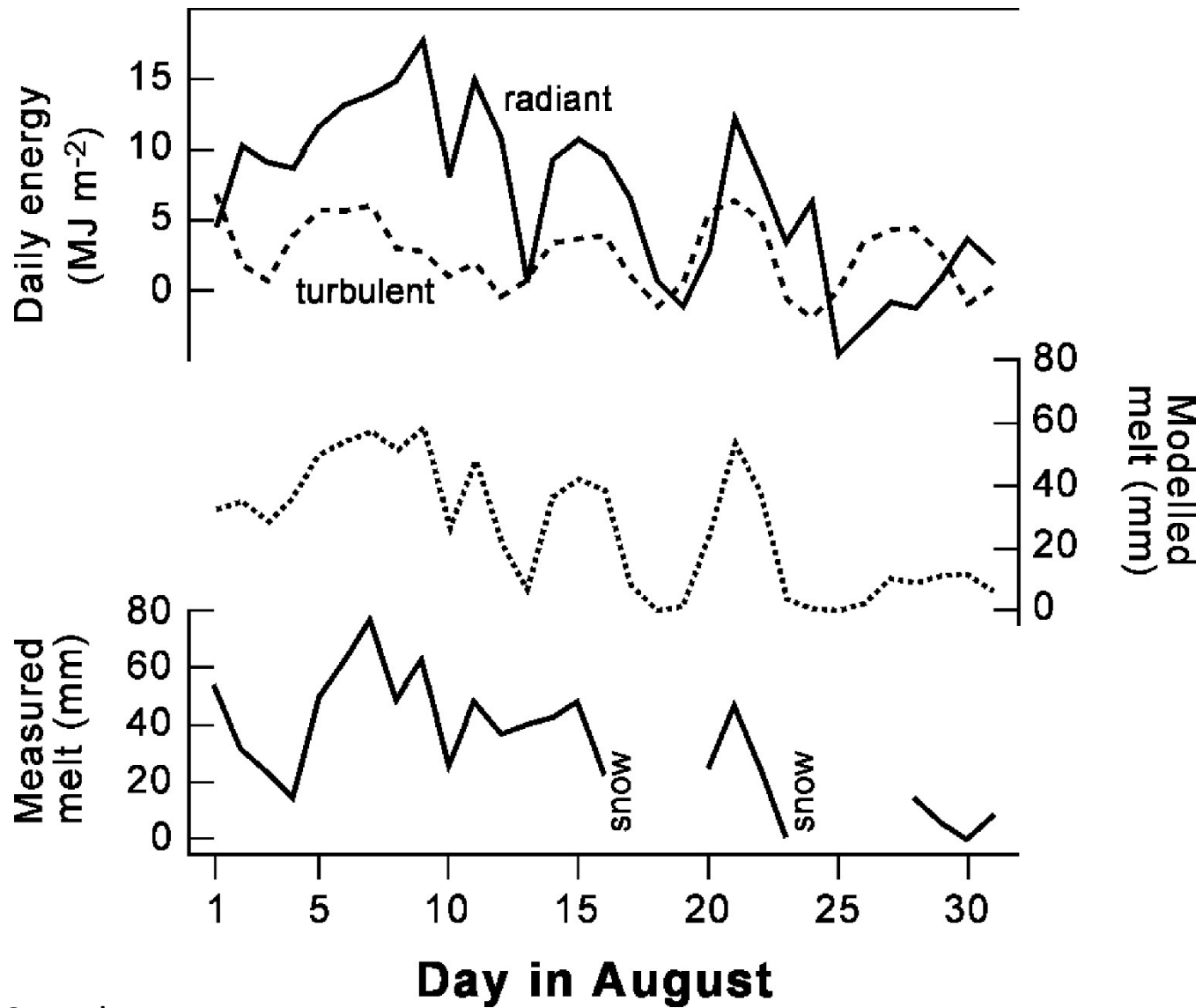
....back to Ablation.

Net energy into the glacier surface = energy used to warm up
the surface layer
+ energy used in melting.

When $T = 0^{\circ}\text{C}$:

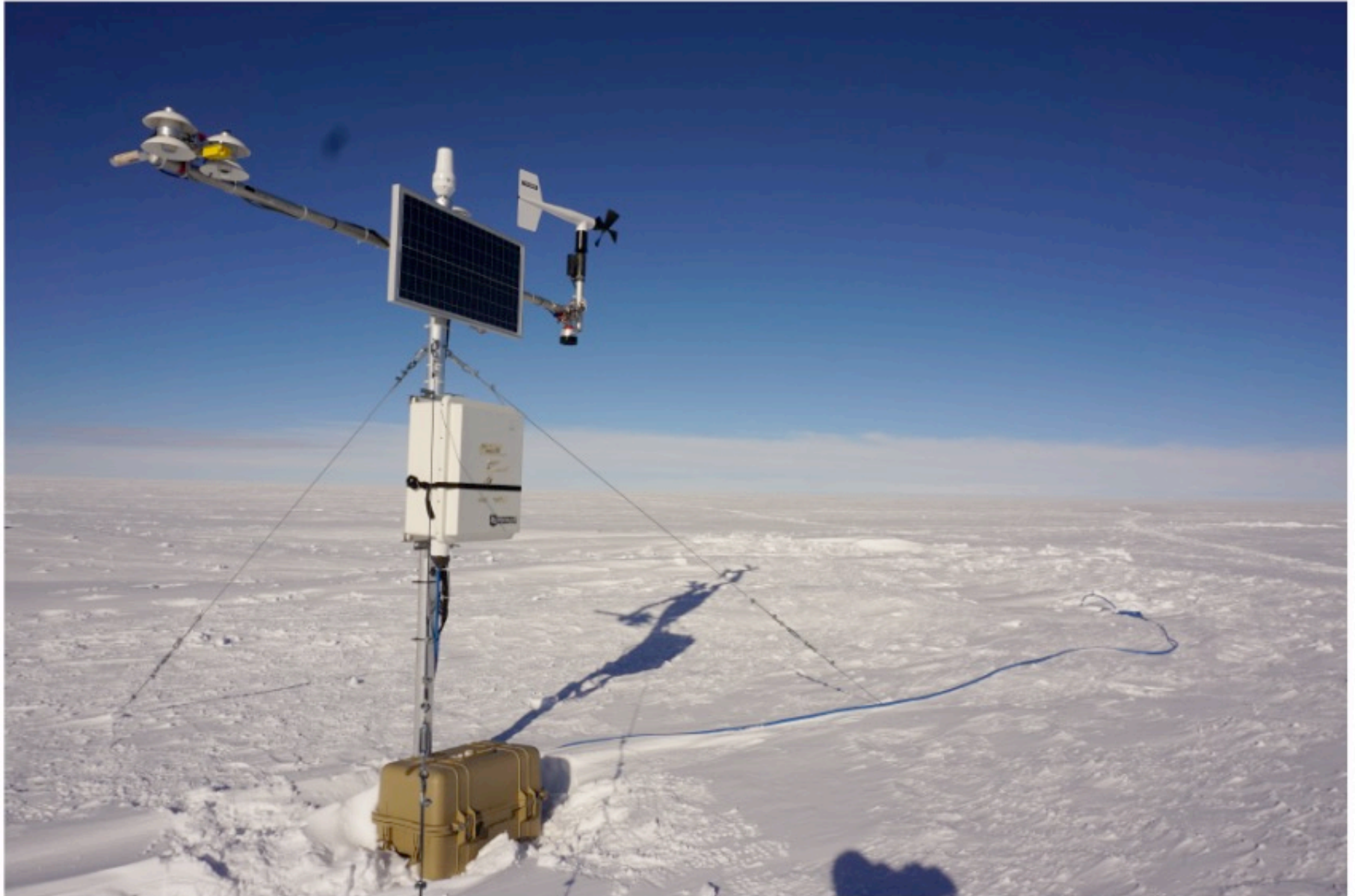
$$E_N = \rho_w L_f \dot{m}$$

$$\dot{m} = \frac{E_N}{\rho_w L_f}$$

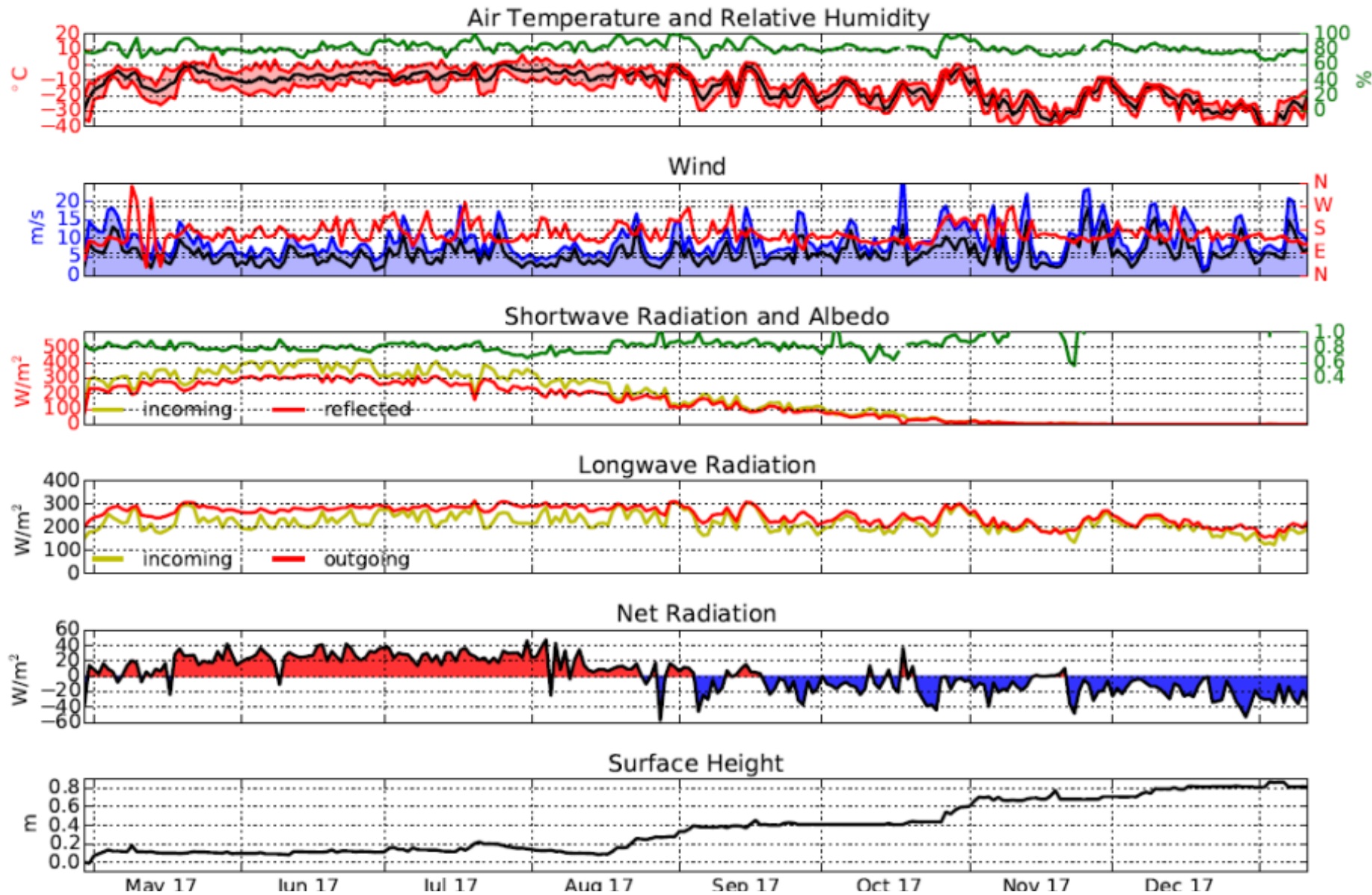


Haig Glacier, Canada,
Cuffey and Paterson (2010)

Automatic weather station (AWS) Greenland Ice Sheet, 2060 m a.s.l.

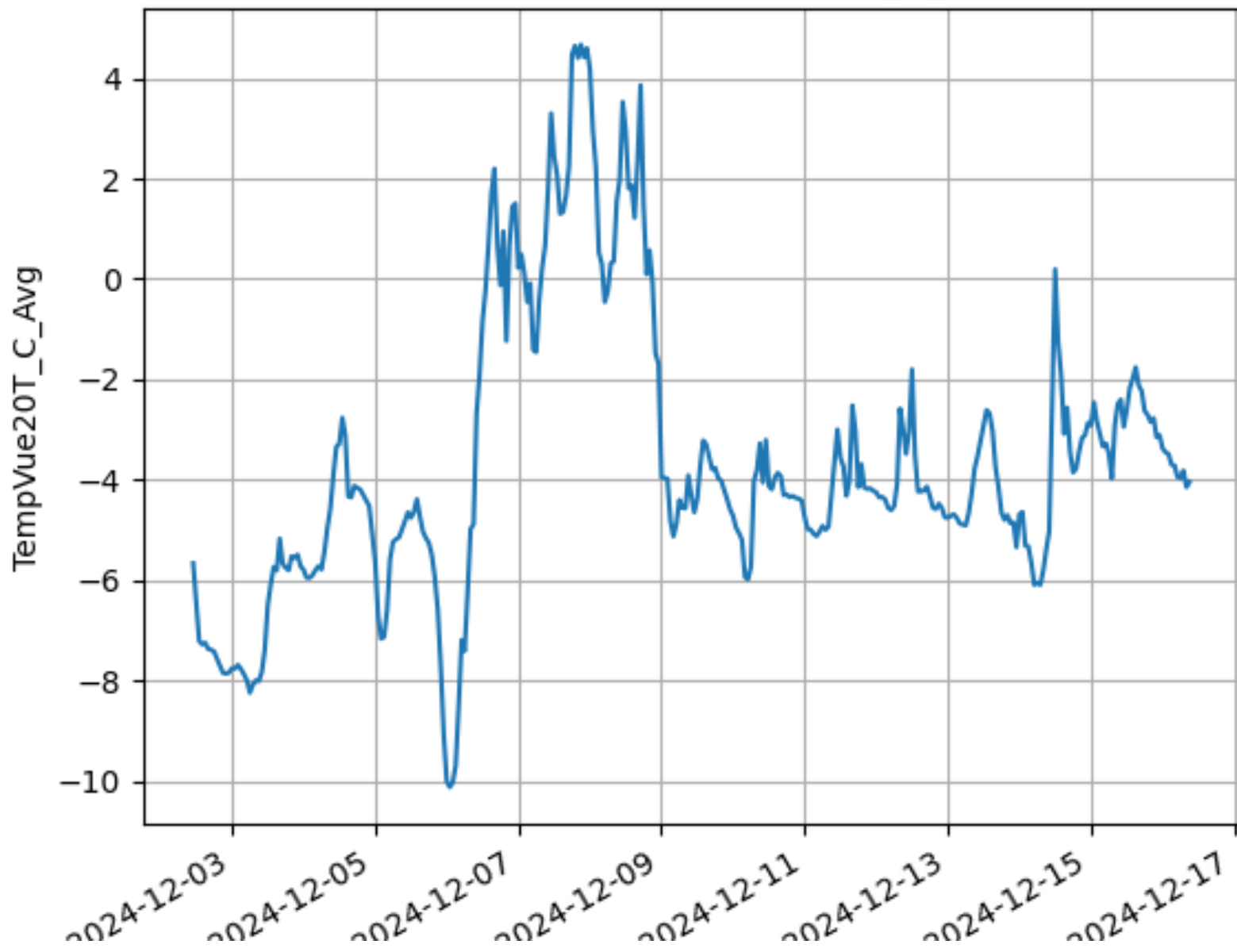


AWS - Site J 2060 m a.s.l.





Time Series of TempVue20T_C_Avg



How do increasing temperatures affect melting and mass balance?

Short-term changes

- Increased sensible heat
- Increased downward longwave radiation
- Early exposure of low albedo ice at the start of the melt season
- Longer melt seasons.

Longer-term changes

- Lowering the surface increases surface temperatures further.
- Increases melt water can enhance ice flow.

How do you measure mass balance?

1. Glaciological method

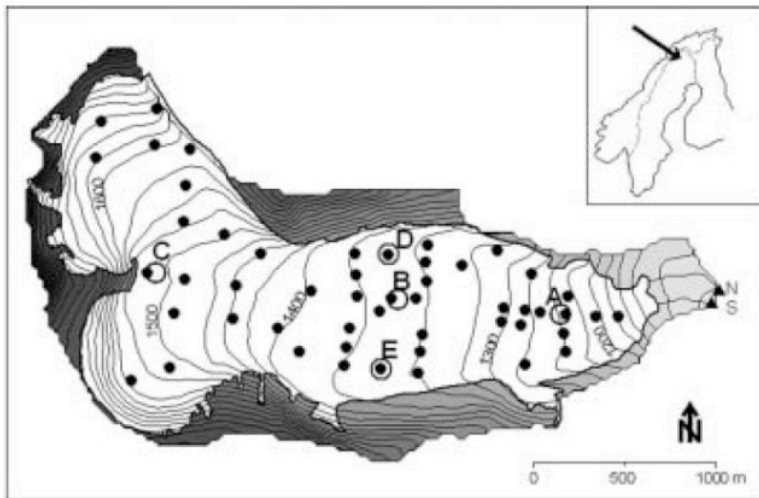
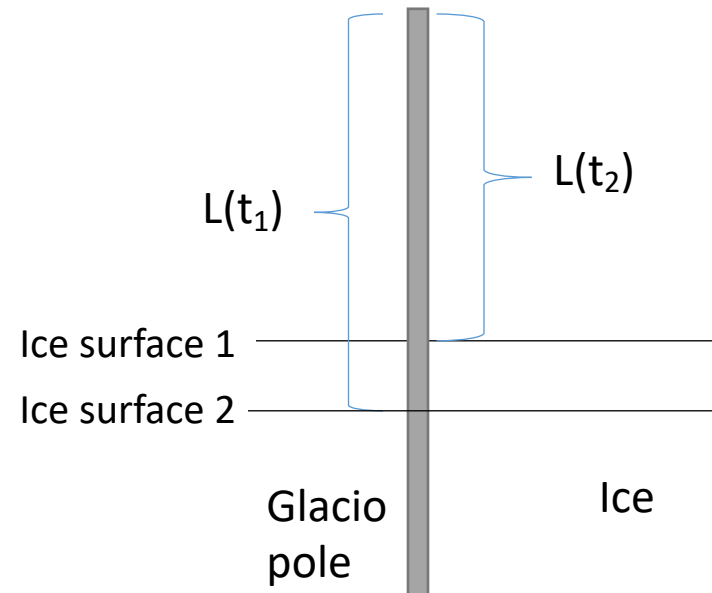


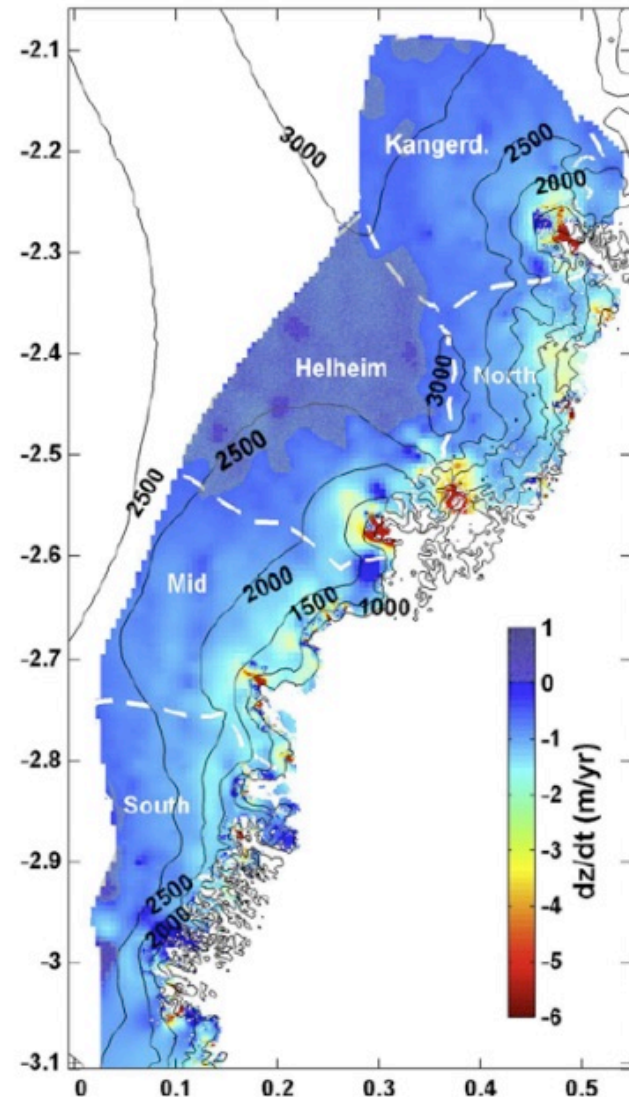
Fig. 1. Drainage basin of Storglaciären with 25 m contour lines. N and S refer to the sites of the water-stage recording stations at the glacial streams Nordjåkk and Sydjåkk. The black dots mark the positions of the ablation stakes in 1994, only slightly differing from those in 1993. The circles labelled A–E denote the sites of five micrometeorological stations on the glacier.



Hock et al. (2005)

How do you measure mass balance?

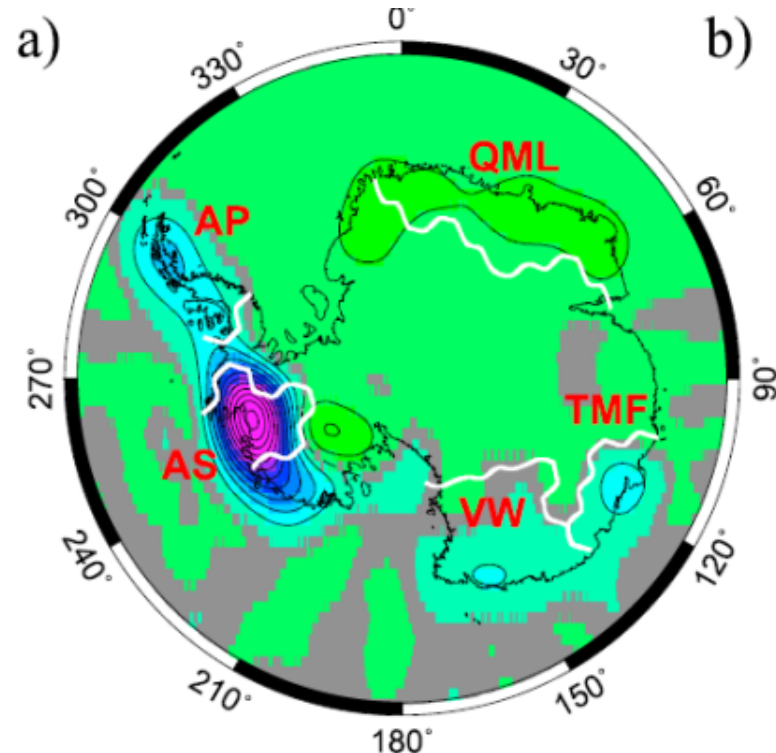
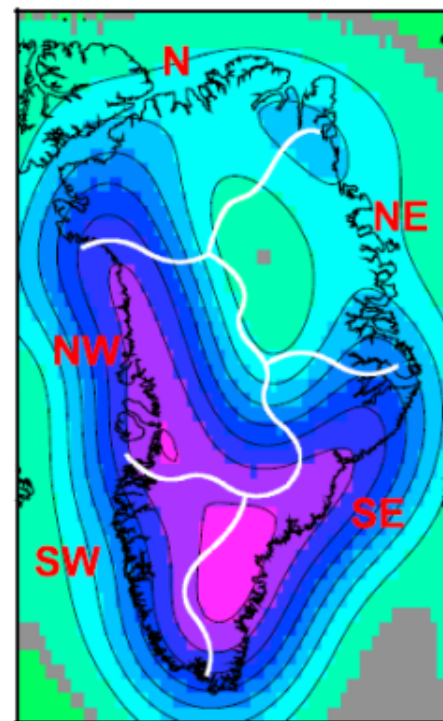
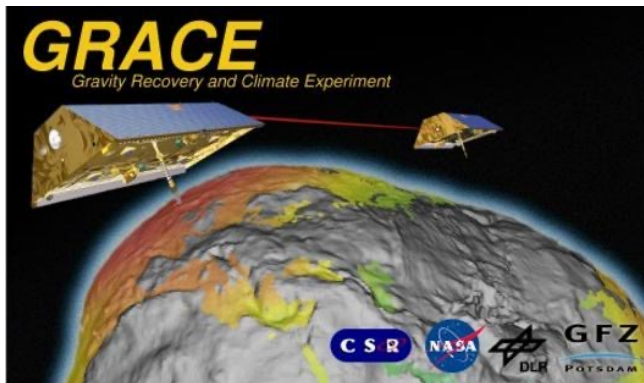
2. Geodetic method



How do you measure mass balance?

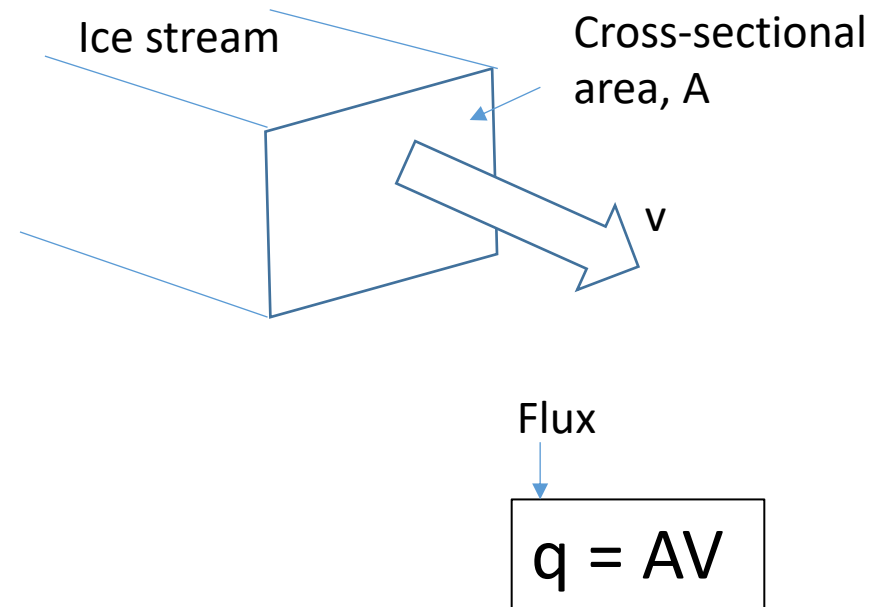
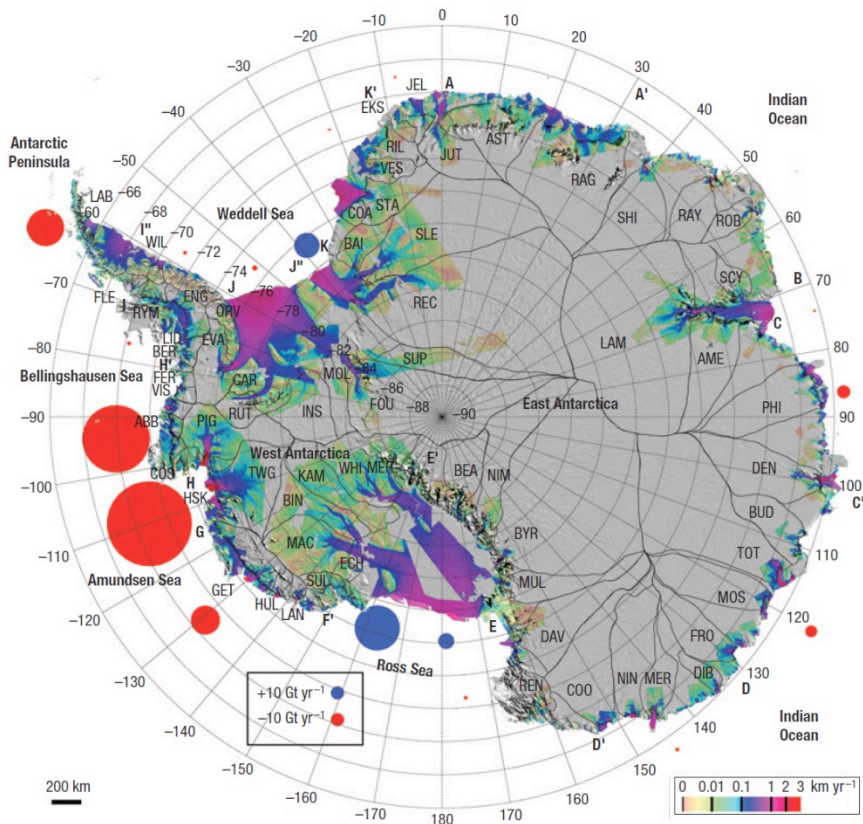
3. Gravimetric method

Velicogna et al. (2014)



How do you measure mass balance?

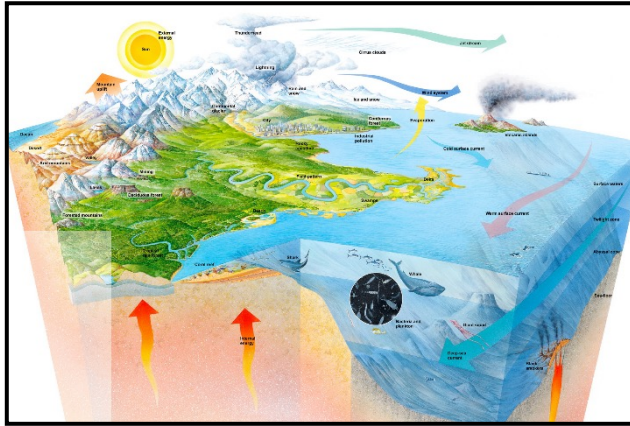
4. Input-output method



Rignot et al. (2008)

Climate Models

The real System



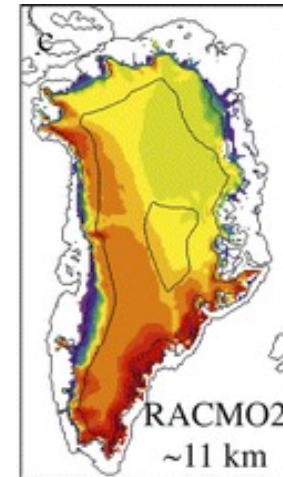
Mathematical model:
e.g. Navier-stokes equations

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \text{ ----- Continuity Equation (1)}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \mathbf{F} + \frac{\mu}{\rho} \nabla^2 \mathbf{u}, \text{ ----- Equations of Motion (2)}$$

Numerical model

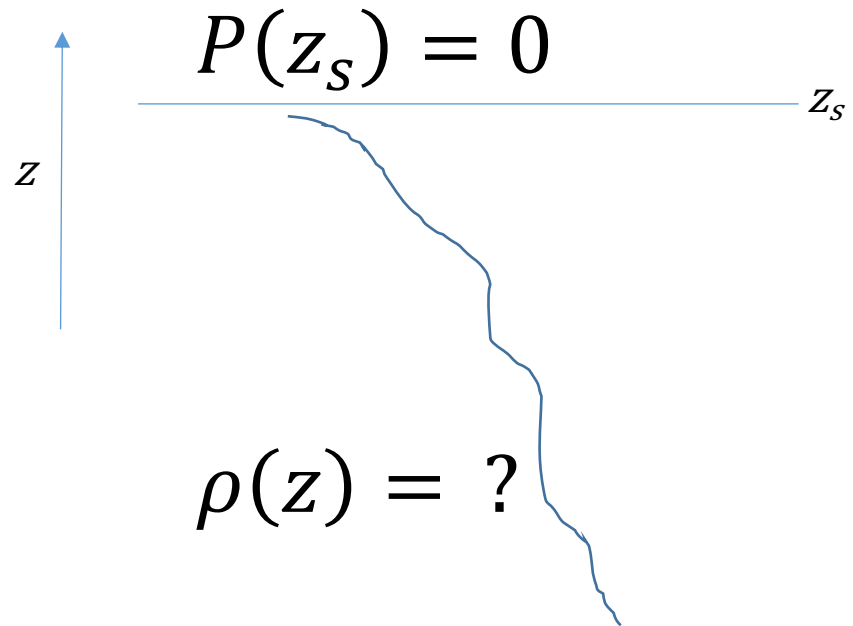
Simulations



Descritization and numerical integration

- Example: integrating the overburden pressure equation

$$\frac{dP}{dz} = \rho(z)g$$



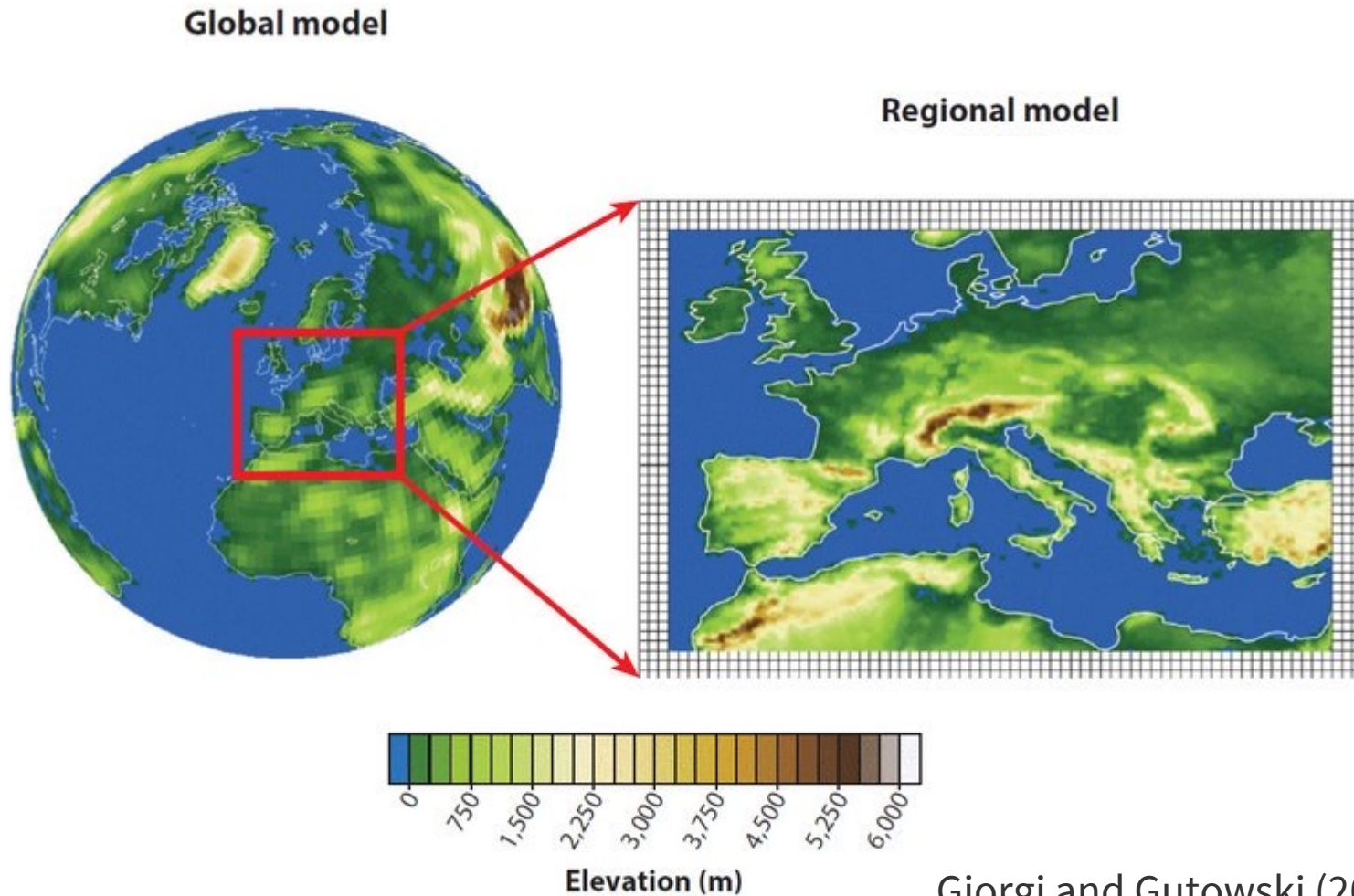
Descritization and numerical integration

Also works in time

$$\frac{dU}{dt} = f(t)$$

$$U(t_0) = U_0$$

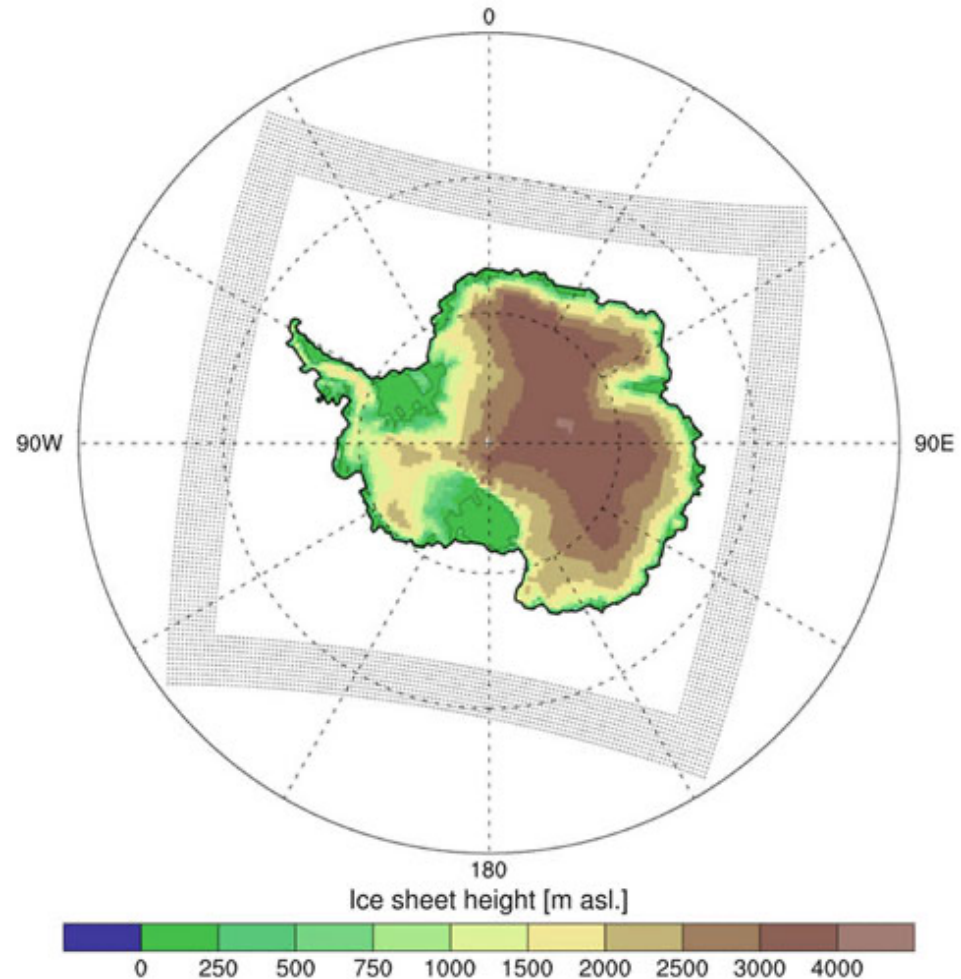
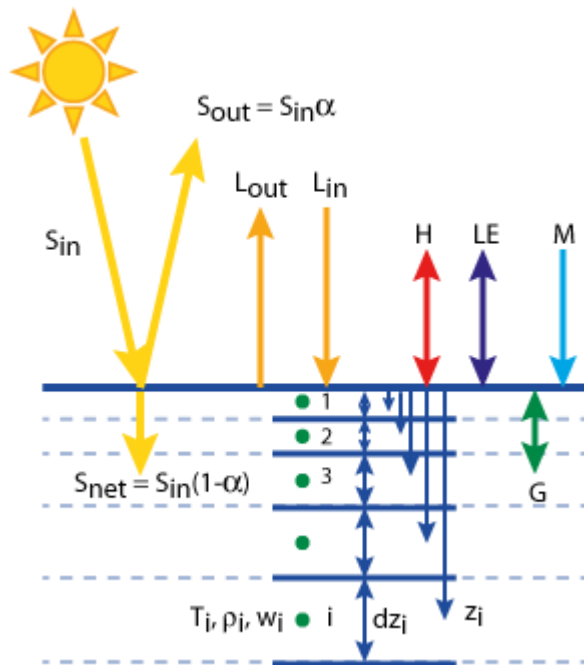
Regional Climate Models



Giorgi and Gutowski (2015)

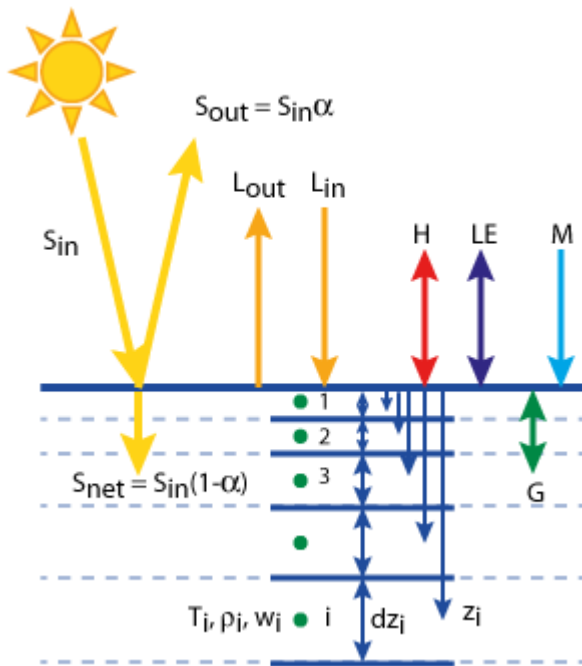
Regional Atmospheric Climate model (RACMO)

Applied to Antarctica:
RACMO/Ant

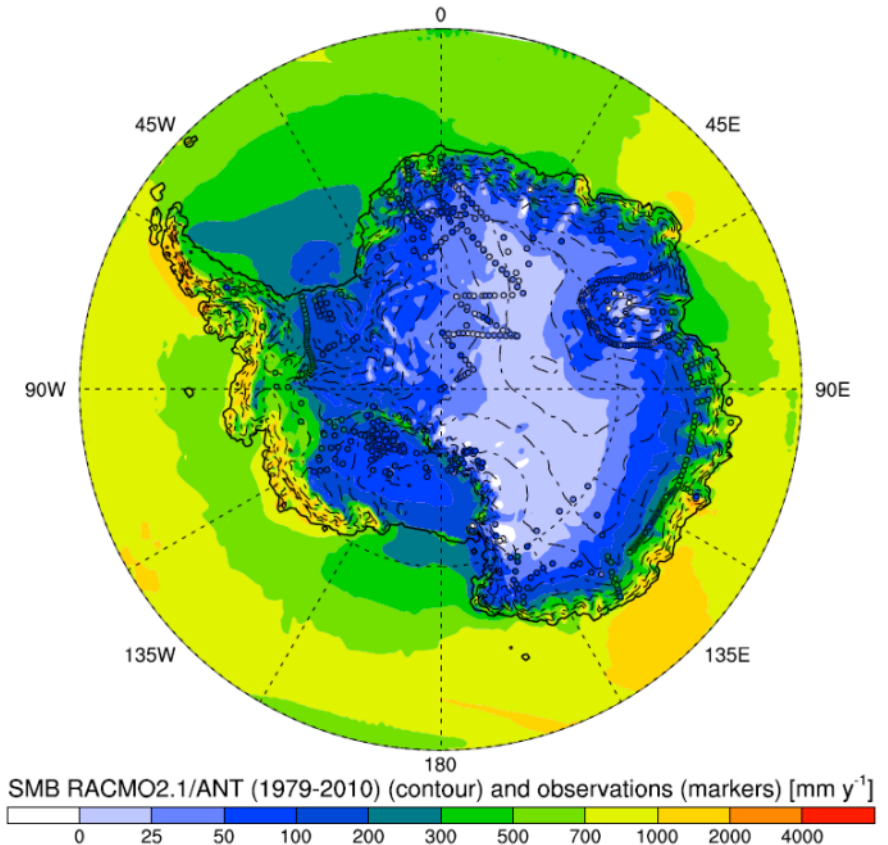


Regional Atmospheric Climate model (RACMO)

Applied to Antarctica:
RACMO/Ant

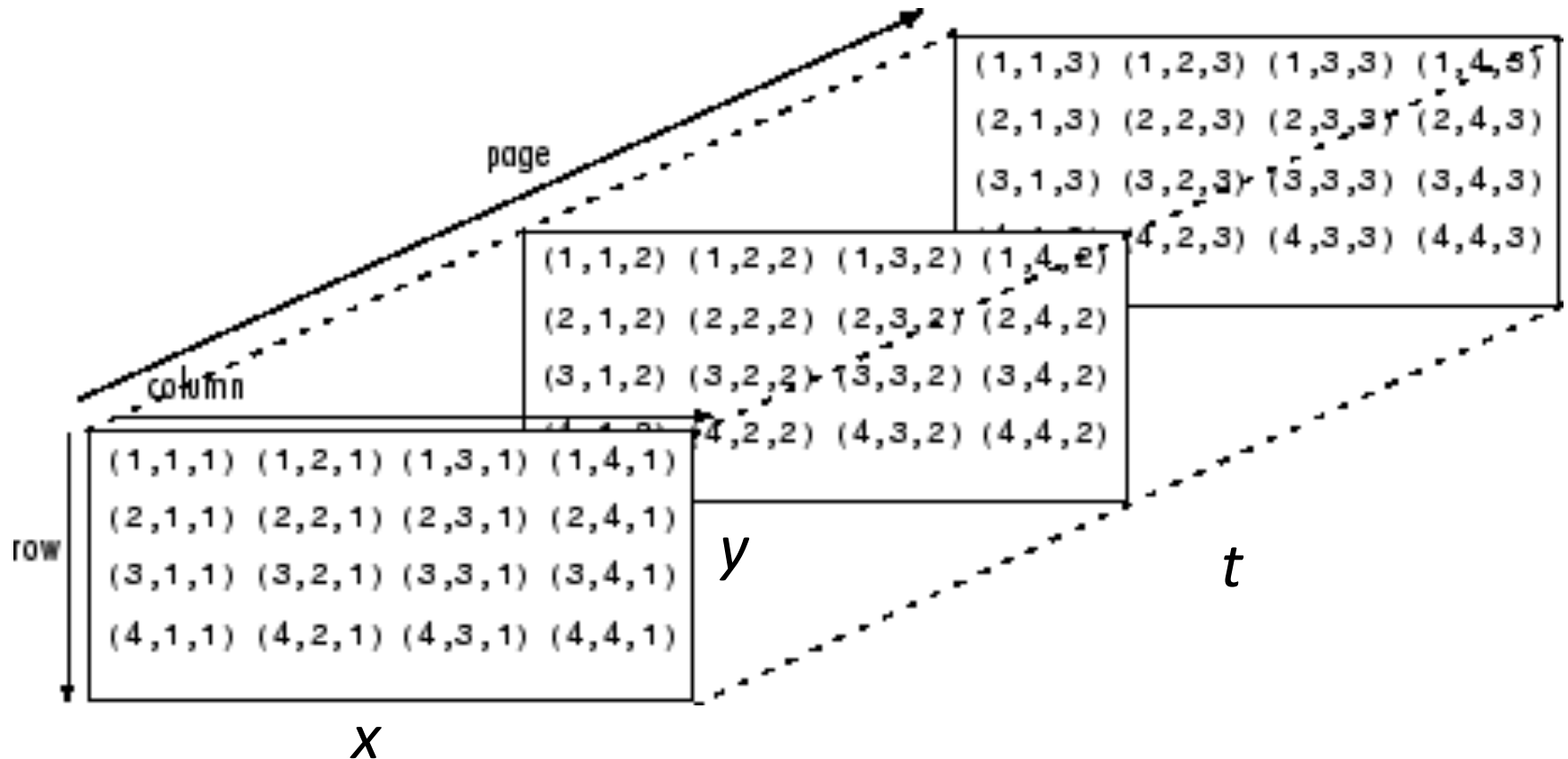


Surface mass balance



Racmo data

2 spatial dimensions + 1 time dimension



On thursday

- We will be manipulating, plotting, analyzing output from the latest version of RACMO

courtesy of Michiel van den Broeke, and Peter Kuipers-Munneke, Utrecht University.

---> read up on multi-dimensional arrays here:

<https://numpy.org/doc/stable/reference/arrays.ndarray.html>

Summary

- Ablation mostly due to calving, basal melt and surface melt.
- Melt is controlled by the surface energy balance. Different components dominate in different places.
- Many interesting phenomena result from variations in energy balance components.

Reading (for Thursday)

1. Section 5.4.5. of Cuffey and Paterson.
2. Chapter 4, up to the end of section 4.2.2 of Benn and Evans

[pdfs on Canvas]

References

Evatt, G.W., Abrahams, I.D., Heil, M., Mayer, C., Kingslake, J., Mitchell, S.L., Fowler, A.C. and Clark, C.D., 2015. Glacial melt under a porous debris layer. *Journal of Glaciology*, 61(229), pp.825-836.

Forbes, J. D. Occasional papers on the theory of glaciers now first collected and chronologically arranged. With a prefatory note on the recent progress and present aspect of the theory, 1809-1868. Edinburgh. A. and C. Black, 1859.

Gourmelen, N., Goldberg, D.N., Snow, K., Henley, S.F., Bingham, R.G., Kimura, S., Hogg, A.E., Shepherd, A., Mouginot, J., Lenaerts, J. and Ligtenberg, S.R., 2017. channelized melting drives thinning under a rapidly melting Antarctic ice shelf. *Geophysical Research Letters*, 44(19), pp.9796-9804.

Kieffer, H.H., Chase, S.C., Martin, T.Z., Miner, E.D. and Palluconi, F.D., 1976. Martian north pole summer temperatures: Dirty water ice. *Science*, 194(4271), pp.1341-1344.

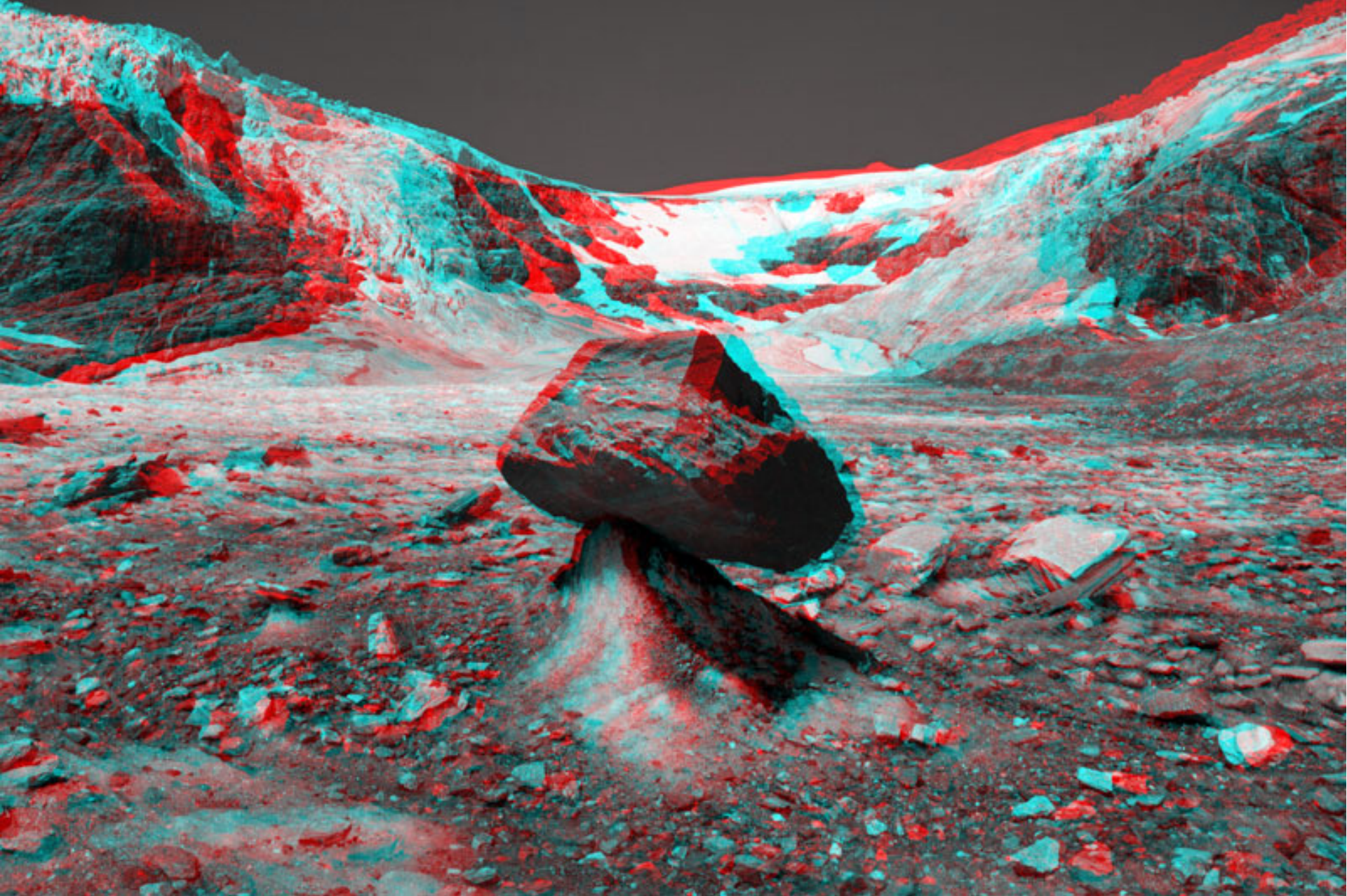
Kingslake, J., Ng, F. and Sole, A., 2015. Modelling channelized surface drainage of supraglacial lakes. *Journal of Glaciology*, 61(225), pp.185-199.

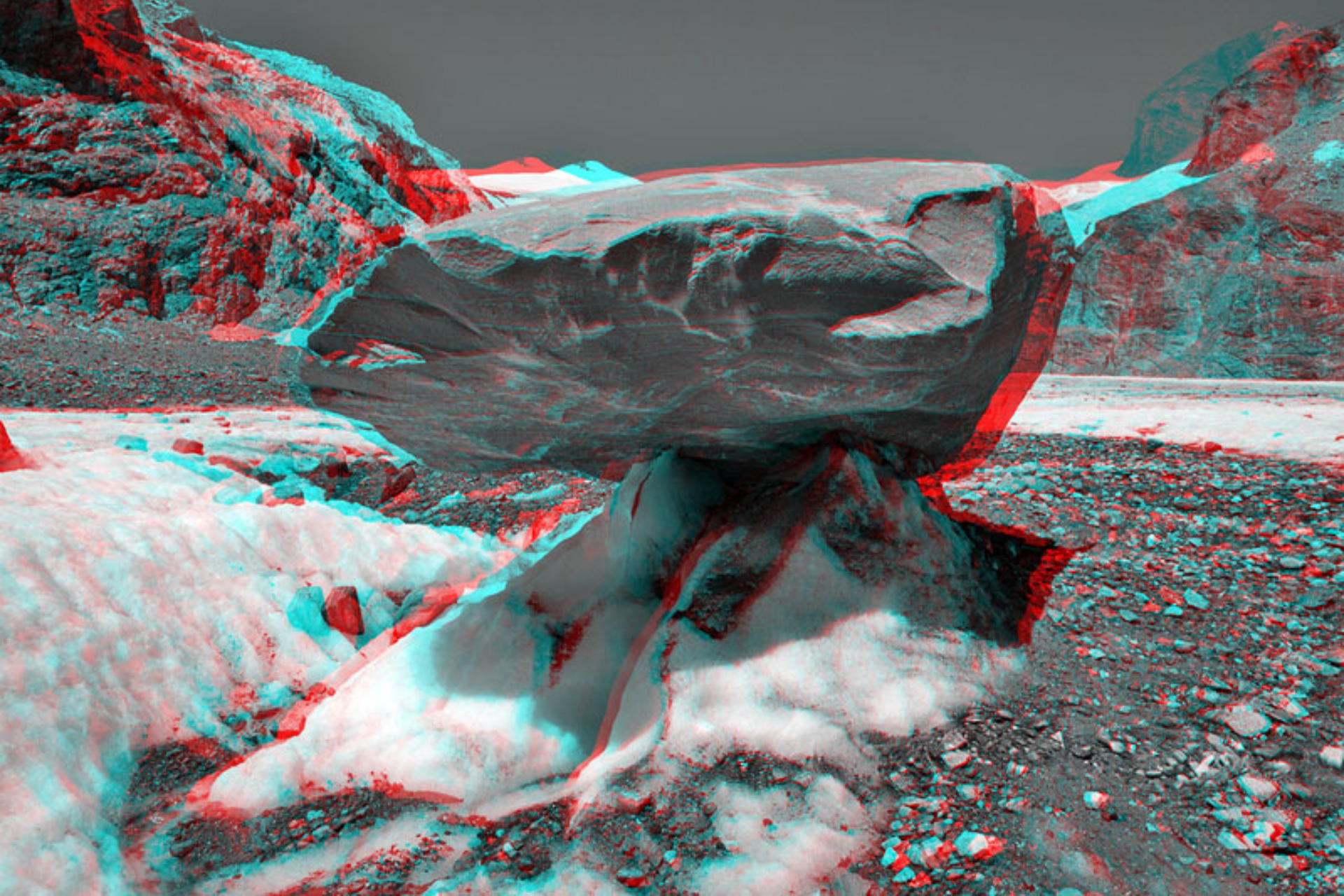
Milkovich, S.M. and Head, J.W., 2005. North polar cap of Mars: Polar layered deposit characterization and identification of a fundamental climate signal. *Journal of Geophysical Research: Planets*, 110(E1).

Nye, J.F., 1958. A theory of wave formation in glaciers. *International Association of Hydrological Sciences*, 47, pp.139-154.

Smith, I.B., Holt, J.W., Spiga, A., Howard, A.D. and Parker, G., 2013. The spiral troughs of Mars as cyclic steps. *Journal of Geophysical Research: Planets*, 118(9), pp.1835-1857.

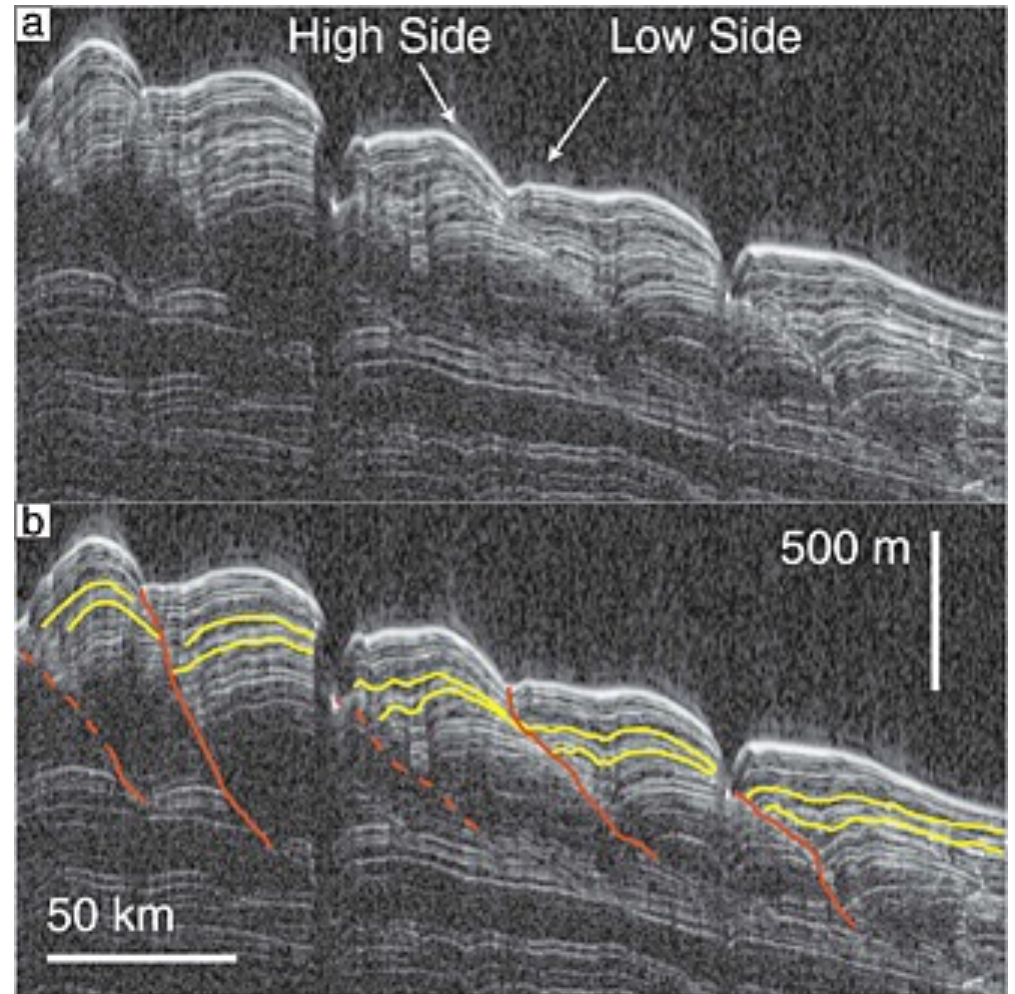
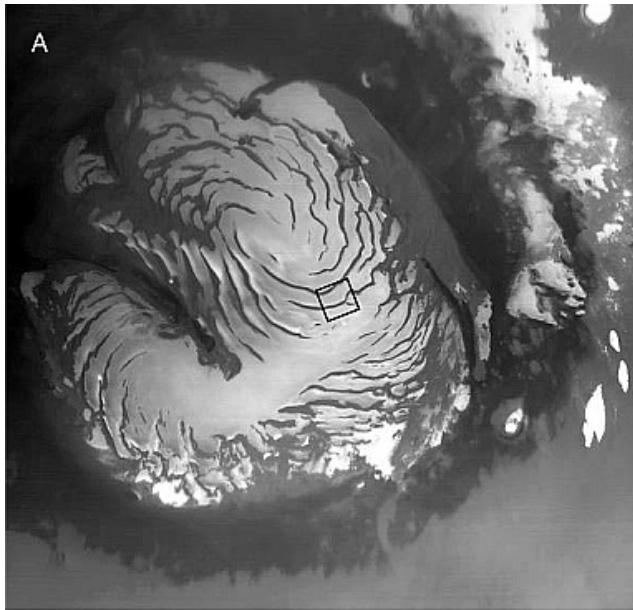
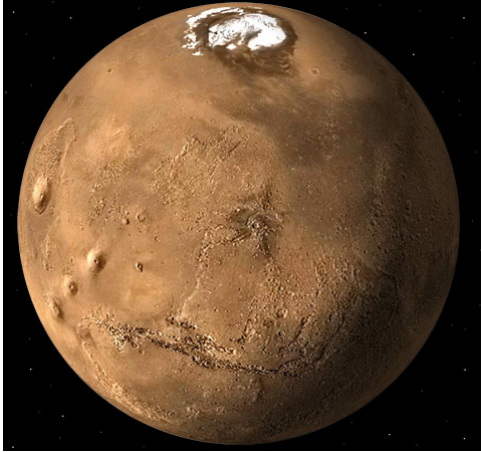
Ablation phenomena 1: Glacier tables

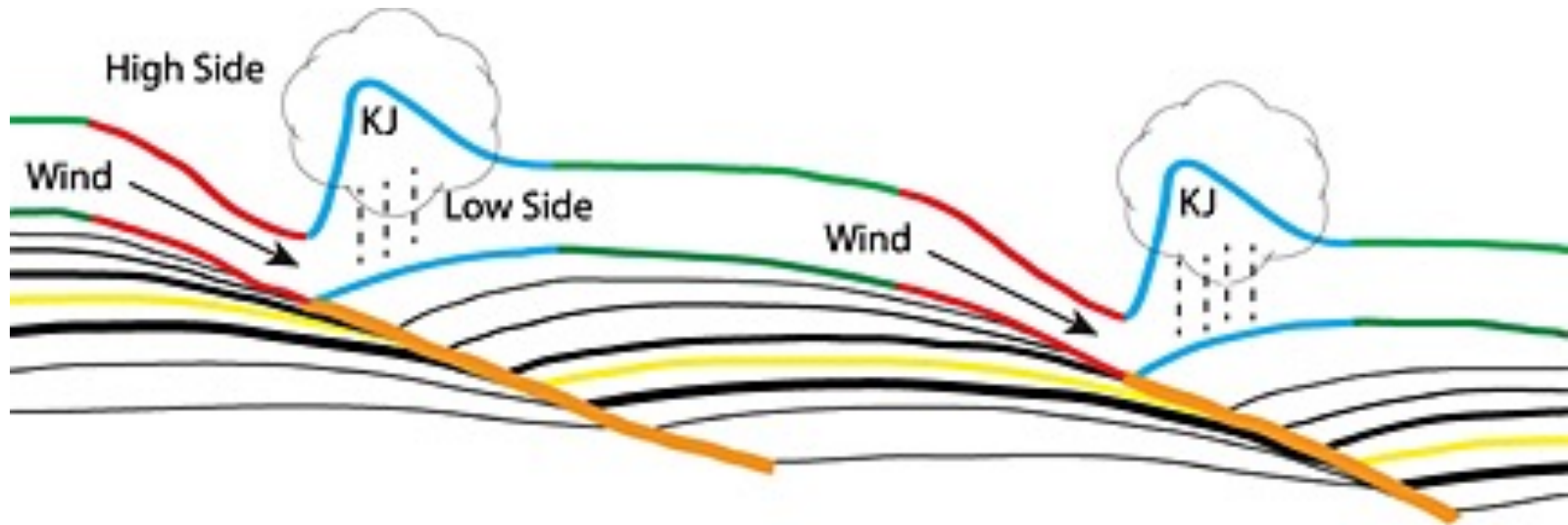




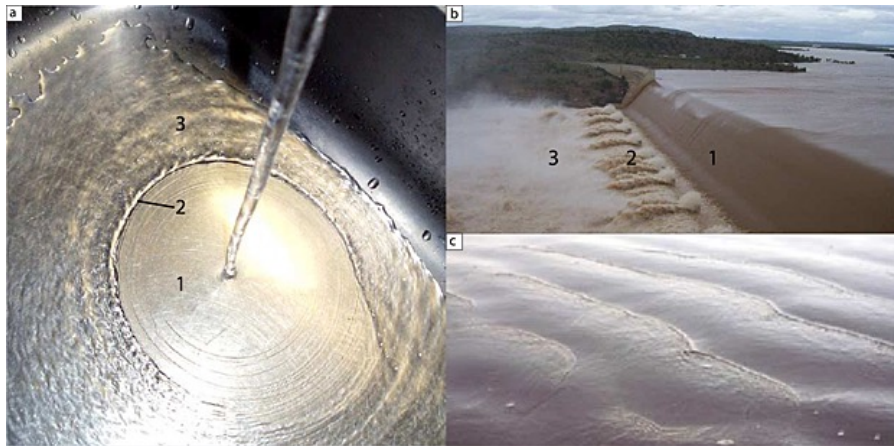
Ablation phenomena 2:

Martian spiral troughs





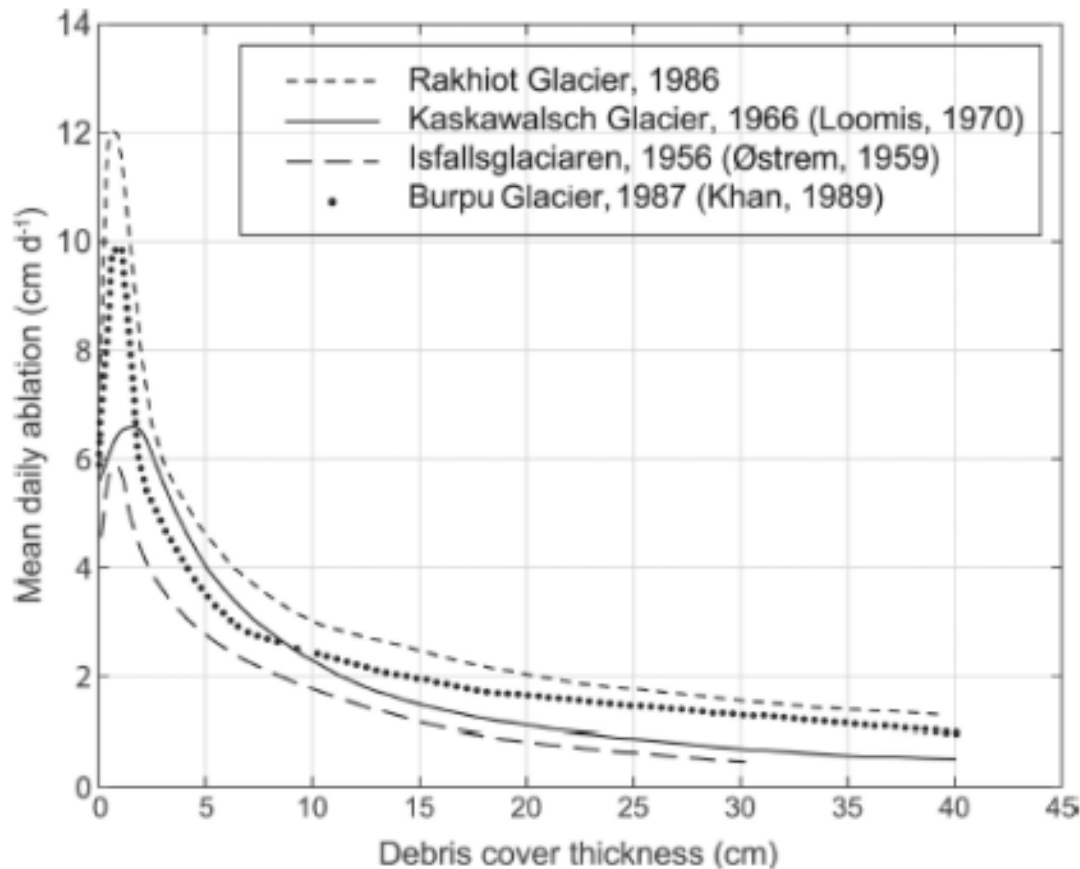
Aeolian counterpart of hydraulic jumps cause erosion on high side and deposition on low side



Smith et al. (2013)

Ablation phenomena 3:

Ostrem Curve



Two explanations for increasing melt rates with thin debris:

1. Decreasing albedo
2. Reduced latent heat flux (Evatt et al., 2015)

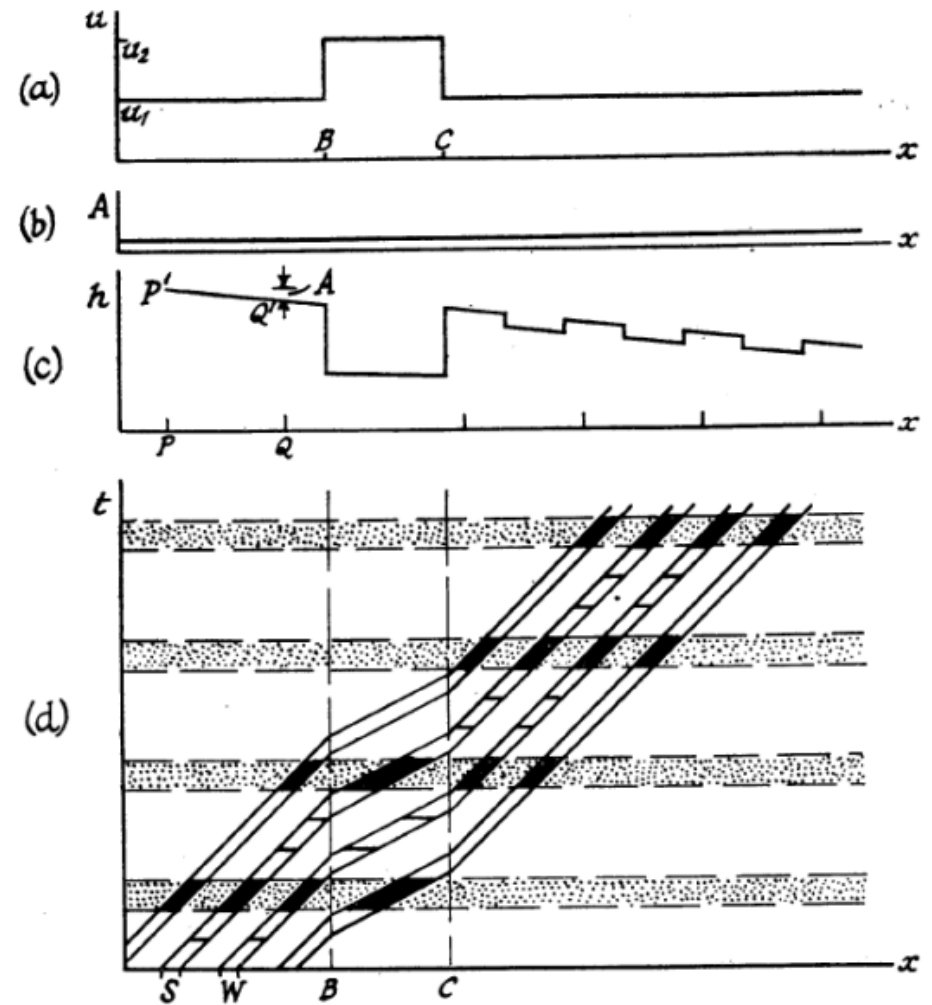
Ablation phenomena 4: Ogives



Austerdalsbreen, Norway



Ablation phenomena 4: Ogives



Nye (1958)